

APPENDIX G

APPLICATION NO. 12/366,378

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	Not yet assigned
Application No.:	Not yet assigned	Group Art Unit:	Not yet assigned
Filed:	February 5, 2009	Examiner:	Not yet assigned

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

This Information Disclosure Statement is submitted:

- ☒ Under 37 CFR § 1.97(b): (1) Within three months of filing of national application; (2) Within three months of date of entry of the national stage as set forth in § 1.491 in an international application; (3) Before mailing of a first Office action on the merits; or (4) Before mailing of a first Office action after the filing of a request for continued examination under § 1.114.
- ☐ Under 37 CFR § 1.97(c): After the period specified in § 1.97(b) but before the mailing date of either a final action under § 1.113 or a notice of allowance under § 1.311, whichever occurs first, and accompanied by either:
- ☐ (1) A statement under § 1.97(e) as set forth below, or
- ☐ (2) A \$180.00 fee under 37 CFR 1.17(p).
- ☐ Under 37 CFR § 1.97(d): After the period specified in § 1.97(c) but on or before payment of the issue fee, and accompanied by:
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Statement under 37 CFR § 1.97(e):

- ☒ Not necessary, because this IDS is filed under § 1.97(b) or § 1.97(c)(2).
- ☐ Each item of information contained in this IDS was cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this IDS; or

- ☐ No item of information contained in this IDS was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the undersigned person after making reasonable inquiry, no item of information contained in this IDS was known to any individual designated in § 1.56(c) more than three months prior to the filing of this IDS.

Form PTO 1449–Information Disclosure Citation is attached with copies of patents, publications, or other information of which Applicants are aware, which may be material to the examination of this application, and for which there may be a duty to disclose in accordance with 37 CFR 1.56. This is not to be construed as a representation that a search has been made or that no better prior art exists, or that a reference is relevant merely because cited.

The Examiner is requested to initial the attached form PTO-1449 and to return a copy of the initialed document to the undersigned as an indication that the attached references have been considered and made of record.

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Respectfully submitted,
DITTHAVONG MORI & STEINER, P.C.

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106

Date: February 5, 2009

Telephone No.: (703) 519-9954

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number			
	Filing Date		2009-02-05	
	First Named Inventor	Karthik RAMANI et al.		
	Art Unit			
	Examiner Name	Not yet assigned		
	Attorney Docket Number	01158-1030		

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	2009-02-05
First Named Inventor	Karthik RAMANI et al.
Art Unit	
Examiner Name	Not yet assigned
Attorney Docket Number	01158-1030

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	2009-02-05
First Named Inventor	Karthik RAMANI et al.
Art Unit	
Examiner Name	Not yet assigned
Attorney Docket Number	01158-1030

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ Fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ None

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Arthur J. Steiner/	Date (YYYY-MM-DD)	2009-02-05
Name/Print	Arthur J. Steiner	Registration Number	26106

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Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

METHOD TO REDUCE MOL DAMAGE ON NiSi

First Named Inventor/Applicant Name:

Karthik Ramani

Filer:

Arthur J. Steiner/Dara Phommachanh

Attorney Docket Number:

NY0281 (01158-1030)

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	330	330
Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1090

Electronic Acknowledgement Receipt

EFS ID:	4738862
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	30671
Filer:	Arthur J. Steiner/Dara Phommachanh
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	NY0281 (01158-1030)
Receipt Date:	05-FEB-2009
Filing Date:	
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Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1090
RAM confirmation Number	2617
Deposit Account	504213
Authorized User	DITTHAVONG,PHOUPHANOMKETH

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NY0281_01158-1030_ADS.pdf	17243 fc5465844603ce578688cc0be2cf0c27d7e8e503	no	2
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2		NY0281_01158-1030_Specifica tion.pdf	101083 1b3ca6a68e8f9b681d65651c00625131096152b5	yes	20
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	15	
	Claims		16	19	
	Abstract		20	20	
Warnings:					
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3	Drawings-only black and white line drawings	NY0281_01158-1030_Drawings .pdf	41594 381f400879c3fe704dd5df267f73309d0e13e953	no	13
Warnings:					
Information:					
4	Oath or Declaration filed	NY0281_01158-1030_Declarati on.pdf	414302 e2a5402509fce23157be284b75922e784e08e49f	no	6
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The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
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5	Information Disclosure Statement Letter	NY0281_01158-1030_IDSTrans mittalLetter.pdf	23457 f5c400b100a4494d53c49ddb4f58a2059585e495	no	2
Warnings:					
Information:					
6	Information Disclosure Statement (IDS) Filed (SB/08)	NY0281_01158-1030_IDS_PTO _SB08.pdf	607940 0b633261f0b096c2cf48bab9d824a7c9d4aa373	no	4
Warnings:					
Information:					

7	Fee Worksheet (PTO-06)	fee-info.pdf	33051 538e0ed43922559182ec80712699e680d9e ebb47	no	2
Warnings:					
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Total Files Size (in bytes):				1238670	
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APPLICATION DATA SHEET

Application Information

Application Type:: Regular
Subject Matter:: Utility
Title:: METHOD TO REDUCE MOL DAMAGE ON
NISI
Attorney Docket Number:: 01158-1030
Total Drawing Sheets:: 13
Small Entity?:: No

Inventor Information

Inventor One Authority Type:: Inventor
Primary Citizenship Country:: India
Status:: Full Capacity
Given Name:: Karthik
Family Name:: RAMANI
City of Residence:: Hopewell Junction
State or Province of Residence:: NY
Country of Residence:: US
Street:: 3603 Chelsea Cove South
City:: Hopewell
State or Province:: NY
Country:: US
Postal or Zip Code:: 12533

Inventor Two Authority Type:: Inventor
Primary Citizenship Country:: US
Status:: Full Capacity
Given Name:: Paul R.
Family Name:: BESSER

City of Residence:: Sunnyvale
State or Province of Residence:: CA
Country of Residence:: US
Street:: 1087 Yorktown Drive
City:: Sunnyvale
State or Province:: CA
Country:: US
Postal or Zip Code:: 94087

Correspondence Information

Correspondence Customer Number:: 30671

Representative Information

Representative Customer Number:: 30671

Assignee Information

Assignee Name:: Advanced Micro Devices, Inc.
Street:: ONE AMD PLACE
P.O. Box 3453
City:: Sunnyvale
State or Province:: CA
Country:: US
Postal or Zip Code:: 94088-3453

Signature:

Signature /Arthur J. Steiner/ Date (YYYY-MM-DD) 2009-02-05

First Name: Arthur J. Last Name: Steiner

Registration Number: 26106

METHOD TO REDUCE MOL DAMAGE ON NiSi

TECHNICAL FIELD

[0001] The present disclosure relates to semiconductor devices with improved nickel silicide layers. The present disclosure is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation.

BACKGROUND

[0002] Conventional semiconductor devices, such as the one shown in FIG. 1, include a refractory metal deposited over all exposed upper surface features of a silicon substrate 101, preferably by means of physical vapor deposition (PVD). The most common refractory metal employed is nickel, which forms a very low resistivity silicide with silicon and does not consume as much Si as other silicides commonly employed. NiSi has a great resistance to etchants which are used to etch silicon oxide, a common insulating material used in semiconductor devices.

[0003] The refractory metal can be sputtered by DC magnetron sputtering in an ultra-high vacuum, multi-chamber system. The formation of an MOS transistor requires the formation of a gate structure 103 and source/drain junctions 105. The gate electrode 103 typically is formed by depositing a layer of heavily doped polysilicon 103a on a metal oxide insulating layer 103b and etching the layers to pattern the electrode. Sidewall spacers 107 are deposited on the opposing surfaces of the patterned gate electrode with oxide liner 109 between spacer 107 and both gate electrode 103 and substrate 101. The sidewall spacers 107 comprise silicon oxide, silicon nitride or silicon oxynitride.

[0004] Generally, in forming the MOS transistor, the refractory metal is deposited after etching the gate electrode 103 and after forming the source/drain junction 105, as illustrated in FIG. 1. After deposition, the refractory metal layer blankets the top surface of the gate electrode 103, sidewall spacers 107, oxide liners

109, and source/drain regions 105. As a result of thermal processing, e.g., a rapid thermal annealing (RTA) process, performed in an inert or reducing atmosphere, the refractory metal reacts with underlying silicon to form electrically conductive silicide layer portions 111 on the top surface of the polysilicon gate electrode 103 and on the exposed surfaces of the substrate where source and drain regions 105 are formed. Unreacted portions of the refractory metal layer, e.g., on the silicon oxide, nitride or oxynitride sidewall spacers 107 and the silicon oxide liner 109 are then removed, as by a wet etching process selective to the metal silicide portions 111.

[0005] Substrates based on “strained silicon” have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption. A layer 101a of silicon-germanium (Si-Ge) is formed on a suitable crystalline substrate, e.g., a Si wafer or a silicon-on-insulator (SOI) wafer 101b. The Si-Ge layer has a greater lattice constant (spacing of Si and Ge atoms therein) relative to the underlying Si. As a consequence, the Si-Ge layer has a compressive strain. Electrons in such compressively strained layers have greater mobility than in conventional Si layers with smaller inter-atom spacings, i.e., there is less resistance to electron flow. For example, electron flow may be up to about 70% faster compared to electron flow in conventional Si. Transistors and IC devices formed with such strained Si layers can exhibit operating speeds up to about 35% faster than those of equivalent devices formed with conventional Si, without necessity for reduction in transistor size. Conventional practices based on strained silicon technology also involve epitaxially growing a relaxed silicon layer on a tensilely stressed silicon layer which is subsequently doped to form relaxed source/drain regions in the relaxed silicon layer.

[0006] Conventional semiconductor devices typically comprise a plurality of active devices in or on a common semiconductor substrate, e.g., CMOS devices comprising at least a pair of PMOS and NMOS transistors in spaced adjacency. The mobility of electrons is faster than the mobility of holes in conventional bulk silicon substrates. Accordingly, in conventional CMOS transistors, the drive current of the PMOS transistor is less than the drive current of the NMOS transistor

creating an imbalance. This imbalance is exacerbated in CMOS transistors fabricated on or within a tensilely stressed active device area formed in a strained lattice semiconductor substrate, e.g., strained Si-Ge on Si, because the increase in electron mobility is greater than the increase in hole mobility.

[0007] Channel carrier mobility of transistors based on strained Si substrates can be increased by applying a stress thereto. In forming P-channel transistors, channel carrier mobility is enhanced by applying a stressed dielectric layer exhibiting high compressive stress for increasing electron mobility. In N-channel transistors, channel carrier mobility is significantly increased by applying a stressed layer exhibiting high tensile stress for increasing hole mobility. Stressed dielectric layers, called stress liners, are applied to transistors wherein the source/drain regions are formed within the strained Si layer, and to transistors having relaxed source/drain regions formed on strained Si layers. The stress liner may comprise silicon carbide, silicon nitride or silicon oxynitride, and may be deposited by plasma enhanced chemical vapor deposition (PECVD) at a thickness of about 200 Å to about 1000 Å. Conventional PECVD conditions may be employed for deposition of a highly compressive layer or highly tensile dielectric layer. In depositing a stress liner exhibiting high compressive stress, both high frequency and low frequency power are applied. When depositing a stressed dielectric layer exhibiting high tensile stress, the low frequency power is significantly reduced.

[0008] Because the stress liners differ for P-channel and N-channel transistors, the stressed dielectric layers are typically patterned by reactive ion etching, which involves physically bombarding the surfaces to subtractively remove material. This etching process involves an “overetch” in order to clear the residue left behind. It is during this overetch process that the underlying NiSi layers are damaged, causing degradation of the NiSi. The degradation can be morphological, wherein mouse bites or missing NiSi is observed, as illustrated in FIG. 1 at 113a-113f, or electrical, wherein the resistivity is increased. Either or both types of degradation ultimately result in poor device parameters such as higher sheet resistance and external resistance.

[0009] A need therefore exists for methodology enabling fabrication of semiconductor devices comprising transistors with degradation resistant NiSi layers and for the resulting improved semiconductor devices having reduced resistance and improved performance.

SUMMARY

[0010] An aspect of the present disclosure is a semiconductor device comprising a transistor with reduced resistance.

[0011] Another aspect of the present disclosure is an efficient method of fabricating a semiconductor device comprising a transistor with reduced resistance.

[0012] Additional aspects and other features of the present disclosure will be set forth in the description which follows and in part will be apparent to those having ordinary skill in the art upon examination of the following or may be learned from the practice of the present disclosure. The advantages of the present disclosure may be realized and obtained as particularly pointed out in the appended claims.

[0013] According to the present disclosure, some technical effects may be achieved in part by a method comprising: forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode; forming a first layer of nickel containing a first amount of platinum over the transistor; forming a second layer of nickel containing a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; and annealing the first and second layers of nickel to form a layer of NiSi over the source/drain regions and the gate electrode, wherein the NiSi layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

[0014] Aspects of the present disclosure include forming the first and second layers of nickel at a combined thickness of about 45 Å to about 100 Å with the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness. Aspects also include annealing to form the layer of NiSi at a thickness of

about 100 Å to about 220 Å. Further aspects include annealing by heating at a temperature of about 340° C to about 450° C, such as at a temperature of about 360° C to about 400° C. This technique can be used to effect annealing in a single step. Another aspect includes annealing by heating at a first stage at a first temperature of about 280° C to about 350° C, for example about 300° C to about 320° C, followed by heating at a second stage at a second temperature of about 360° C to about 450° C, for example about 380° C to about 420° C. Additional aspects include forming the first layer of nickel with a platinum content of about 8 weight % (“wt. %”) to about 12 wt. %, for example about 10 wt. %, and the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %, for example about 30 wt. % to about 50 wt. %. Aspects also include forming a capping layer of titanium nitride over the second layer before annealing, and removing the capping layer after annealing, followed by etching to remove unreacted nickel.

[0015] Another aspect of the present disclosure is a semiconductor device comprising: a silicon substrate; a transistor, comprising source/drain regions and a gate electrode, on the substrate; and a NiSi layer on the gate electrode and source/drain regions, the NiSi layer comprising platinum having a composition gradient, the platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

[0016] Aspects include devices with a P-channel transistor, an N-channel transistor, a first NiSi layer on the P-channel transistor, a second NiSi layer on the N-channel transistor, the first and second NiSi layers each comprising platinum having a concentration gradient with platinum increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors, a first stress liner on the P-channel transistor, and a second stress liner on the N-channel transistor. Further aspects include the NiSi layers having a thickness of about 100 Å to about 220 Å, for example about 100 Å to about 150 Å.

[0017] Another aspect of the present disclosure is a method of fabricating a semiconductor device, the method comprising: forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate

electrode and source/drain regions formed in the silicon substrate; depositing a first layer of nickel containing a first amount of platinum on both transistors; depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; depositing a titanium nitride capping layer on the second layer of nickel, annealing the first and second layers of nickel to form a single layer of NiSi containing platinum having a concentration gradient with the platinum increasing in amount in a direction away from the gate electrode and source/drain regions; etching to remove the capping layer and unreacted nickel; forming a first stress liner exhibiting high compressive stress over both transistors; selectively removing the first stress liner from the N-channel transistor; forming a second stress liner exhibiting high tensile stress over both transistors; and selectively removing the second stress liner from the P-channel transistor.

[0018] Additional aspects and technical effects of the present disclosure will become readily apparent to those skilled in the art from the following detailed description wherein embodiments of the present disclosure are described simply by way of illustration of the best mode contemplated to carry out the present disclosure. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawing and in which like reference numerals refer to similar elements and in which:

[0020] FIG. 1 schematically illustrates a conventional transistor device with a degraded NiSi layer;

[0021] FIGS. 2 through 6 schematically illustrate sequential steps of a method in accordance with an embodiment of the present disclosure;

[0022] FIGS. 7 through 15 schematically illustrate sequential steps of a method in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0023] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of exemplary embodiments. It should be apparent, however, that exemplary embodiments may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring exemplary embodiments.

[0024] The present disclosure addresses and solves the problem of NiSi layer degradation during subsequent processing, as during middle of the line processing to remove overlying stress liners, which NiSi layer degradation causes resistance problems and adversely affects transistor performance. In accordance with embodiments of the present disclosure, a transistor device is provided having a NiSi layer formulated to resist degradation, advantageously using methodology that does not impact process flow. In accordance with embodiments of the present disclosure, NiSi layer damage is prevented or significantly reduced by formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode, thereby shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.

[0025] Embodiments of the present disclosure include forming two layers of nickel on a silicon substrate and a silicon gate electrode, the first layer having a relatively low concentration of platinum and the second having a relatively high concentration of platinum, and annealing the two layers to form a single layer of NiSi. During annealing, platinum diffuses to form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode. It was found that such a concentration gradient of platinum effectively

shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.

[0026] Methodology in accordance with embodiments of the present disclosure includes forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode, depositing a first layer of nickel containing a relatively low amount of platinum, for example about 8 wt. % to about 12 wt. %, e.g., about 10 wt. %, over the transistor, forming a second layer of nickel containing a relatively high amount of platinum, such as about 30 wt. % to about 70 wt. %, e.g., about 30 wt. % to about 50 wt. %, on the first layer. The first and second layers of nickel are formed at a combined thickness of about 45 Å to about 100 Å, with the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness. A capping layer of silicon nitride is formed over the second nickel layer. An RTA process is then implemented to convert the platinum containing nickel layers into a single layer of NiSi over the source/drain regions and the gate electrode, such that the NiSi layer has a platinum composition gradient with the amount of platinum increasing in a direction away from the source/drain regions and the gate electrode. The resulting platinum concentration gradient protects the NiSi film without increasing the resistivity of the layer.

[0027] The annealing process may be controlled to form a layer of NiSi at a thickness of about 100 Å to about 220 Å. Annealing may comprise a single heating stage at a temperature of about 340° C to about 450°C, such as at a temperature of about 360° C to about 400° C. Alternatively, a two stage annealing process can be performed, as by heating at a first temperature of about 280° C to about 350° C, such as about 300° C to about 320° C, followed by heating at a second temperature of about 360° C to about 450°C, such as about 380° C to about 420° C. In accordance with embodiments of the present disclosure, the capping layer is removed after annealing followed by etching to remove unreacted nickel, employing conventional methodology.

[0028] Embodiments of the present disclosure include forming two transistors, one P-channel and one N-channel transistor. A first stress liner exhibiting high compressive stress can be formed over both transistors, and the first

stress liner is selectively removed, as by etching, from the N-channel transistor. A second stress liner exhibiting high tensile stress can be subsequently formed over both transistors, and the second stress liner is selectively removed, as by etching, from the P-channel transistor. During etching of the stress liners, the thus formed platinum concentration gradient prevents the NiSi from being etched and damaged.

[0029] Still other aspects, features, and technical effects will be readily apparent to those skilled in this art from the following detailed description, wherein preferred embodiments are shown and described, simply by way of illustration of the best mode contemplated. The disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

[0030] A method in accordance with an embodiment of the present disclosure is schematically illustrated in FIGS. 2 through 5, wherein like reference characters denote like features. Adverting to FIG. 2, strained Ge-Si layer 201 is formed over Si layer 203. Ge-Si layer 201 may be globally strained or locally strained under the source/drain regions. Gate electrode 205 is formed over strained Ge-Si layer 201 with a gate dielectric 207 therebetween. Lightly doped source/drain regions 213a are formed in Ge-Si layer 201 using the gate electrode 205 as a mask in a conventional manner. A sidewall spacer comprising a silicon oxide liner 209, as at a thickness of about 60 Å to about 600 Å, is formed on side surfaces of the gate electrode 205 and on a portion of the upper surface of strained Ge-Si layer 201. Silicon nitride spacers 211 are then formed on the silicon oxide liner 209 as by employing PECVD followed by etching.

[0031] It should be appreciated that liner 209 may be deposited by ALD and may also comprise silicon nitride. Silicon oxide liner 209 advantageously prevents consumption of the gate electrode 205 by silicidation on the side surfaces thereof, and advantageously prevents a subsequently formed thin layer of NiSi on the silicon nitride sidewall spacers 211 from contacting a NiSi contact layer on the upper surface of the gate electrode 205 and/or from contacting NiSi contact layers

on the upper surface of the strained Si layer 201, thereby preventing NiSi bridging along the silicon nitride sidewall spacers.

[0032] Subsequently, heavily doped source/drain regions 213b are formed in Ge-Si layer 201 using gate electrode 205 and sidewall spacers 209 and 211 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

[0033] Adverting to FIG. 3, a first layer of nickel 301 is deposited over the source/drain regions 213b, the spacers 211, the oxide liner 209, and the gate electrode 205. First layer of nickel 301 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. The first layer of nickel can be formed by sputtering using a target comprising Ni and 10 wt. % Pt. Over the first layer of nickel 301, a second layer of nickel 303 is deposited. The second layer of nickel 303 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The second layer of nickel can be sputtered using a target of Ni containing an amount of Pt within the range of 30 to 100 %. The first nickel layer 301 and second nickel layer 303 may be formed to a combined thickness of about 45 Å to about 100 Å using a conventional deposition tool, for example on an AMAT Endura sputtering system, at a power of 1500W to 2500W, e.g., 2000W, and using an Argon flow of 20sccm to 40sccm, e.g., 2000sccm, or an Argon and Hydrogen mixture with a flow of 3sccm to 10sccm, e.g., 5sccm. The first layer of nickel 301 may be formed at a thickness of about 60 % to about 80 % of the combined thickness, such as about 70 % of the combined thickness.

[0034] A capping layer 305 may be formed over the second nickel layer 303 to prevent oxidation of the underlying metal. Capping layer 305 may be titanium nitride.

[0035] RTA and etch processes are performed to form a NiSi layer 401 over the source/drain regions 213b and the gate electrode 205, as illustrated in FIG. 4. The resulting NiSi layer 401 has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the

opposite surface. NiSi layer 401 may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

[0036] Nickel silicide is commonly doped with platinum to improve the material and electrical properties. Platinum has proven to be effective in preventing the nickel disilicide formation (NiSi₂). It was found that strategic platinum doping can also be used to effectively shield a NiSi layer from physical damage during MOL processing (post NiSi processing) while concomitantly maintaining the NiSi integrity. However, platinum also increases the resistivity of NiSi. In accordance with embodiments of the present disclosure, effective shielding of a NiSi layer is achieved while avoiding a problematic increase in resistivity by compositionally grading a platinum doped NiSi layer.

[0037] The RTA process may be a one-step or two-step process (with a clean in between the two steps) in an inert or reducing atmosphere, e.g., a nitrogen (N₂)-containing atmosphere. The single step anneal may be implemented at a temperature of about 340° C to about 450° C, for example at a temperature of about 360° C to about 400° C, for about 5 to 35 seconds. During the single step anneal the nickel layers are converted directly to a low resistivity monosilicide (NiSi).

[0038] The two-step anneal may be conducted by first heating at a temperature of about 280° C to about 350° C, for example about 300° C to about 320° C, e.g., 300° C, during which metal rich silicide, Ni₂Si, is formed. During the second step, the temperature may be raised to about 360° C to about 450° C, such as about 380° C to about 420° C, e.g., 420° C. During the second annealing step, the metal rich silicide is converted to monosilicide. The total time for the two step annealing can be the same as for the single step anneal process, for example 5 seconds for the first step and 30 seconds for the second step. The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough interfaces between the NiSi and the underlying Si layer.

[0039] After NiSi layer 401 is formed, the capping layer 305 and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are

removed by an anisotropic sputter etching or by immersing the wafer into a solution of H_2SO_4 , H_2O_2 , and water (SPM) or a solution of NH_4OH , H_2O_2 and water (APM).

[0040] As illustrated in FIG. 5, the liner 209 and sidewall spacers 211 are then removed, exposing a portion of the upper surface of strained Si layer 201 between the silicide layers 401 and the side surfaces of gate electrode 205, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer. A highly stressed dielectric layer 601, such as a silicon nitride layer exhibiting high compressive stress, is then deposited by PECVD, as shown in FIG. 6. The highly stressed dielectric layer 601 serves to increase channel electron mobility, thereby increasing the drive current.

[0041] Another embodiment of the present disclosure is schematically illustrated in FIGS. 7 through 15. Adverting to FIG. 7, a CMOS device is schematically illustrated comprising an NMOS transistor portion at the left and a PMOS transistor portion on the right, wherein similar features are denoted by similar reference characters. Strained Si layer 701 is formed over Si-Ge layer 703. As in previous discussed embodiments, Si layer 701 may be globally strained or locally strained in the source/drain regions. Gate electrodes 705 are formed over strained Si layer 701 with gate dielectric 707 therebetween. Lightly doped source/drain regions 713a are formed in Si layer 701 using gate electrodes 705 as a mask in a conventional manner. Sidewall spacers comprising oxide liner 709, as at a thickness of about 60 Å to about 600 Å, are formed on side surfaces of gate electrodes 705 and on portions of the upper surface of strained Si layer 701. Silicon oxide liner 709 can be formed in the same manner as discussed with respect to silicon oxide liner 209 in FIG. 2. Silicon nitride spacers 711 are then formed on silicon oxide liners 709 as by employing PECVD followed by etching.

[0042] Subsequently, heavily doped source/drain regions 713b are formed in Si layer 701 using gate electrodes 705 and sidewall spacers 709 and 711 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

[0043] As illustrated in FIG. 8, a first layer of nickel 801 is deposited over the source/drain regions 713b, spacers 711, oxide liners 709, and gate electrodes 705.

First layer of nickel 801 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. Over the first layer of nickel 801, a second layer of nickel 803 is deposited. Second layer of nickel 803 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The first nickel layer 801 and second nickel layer 803 may be formed in a conventional deposition environment, for example on an AMAT Endura sputtering system, to a combined thickness of about 45 Å to about 100 Å. The first layer of nickel 801 may be formed at a thickness of about 60 % to about 80 % of the combined thickness, such as about 70 % of the combined thickness.

[0044] A capping layer 805 is formed over the second nickel layer 803 to prevent oxidation of the underlying metal. Capping layer 805 may be titanium nitride.

[0045] RTA and etch processes are performed to form NiSi layers 901 over source/drain regions 713b and gate electrodes 705, as illustrated in FIG. 9. The resulting NiSi layers 901 have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the bottom surface that contacts the silicon gate electrodes 705 and source/drain regions 713b, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layers 901 may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å. The RTA process may be a one-step or two-step process, as described above with respect to FIG. 4.

[0046] After NiSi layers 901 are formed, the capping layer 805 and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

[0047] As illustrated in FIG. 10, liners 709 and sidewall spacers 711 are then removed exposing portions of the upper surface of strained Si layer 701 between the silicide layers 901 and the side surfaces of gate electrodes 705, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer.

[0048] A highly compressive stressed silicon nitride film 1101, e.g., having a compressive stress greater than 1.5 GPa, is then deposited over both the NMOS and PMOS transistors as shown in FIG. 11. Deposition of highly compressive stressed silicon nitride film 1101 may be implemented in a conventional manner. Multiple layer deposition and plasma treatment further increases compressive stress. Subsequently, a thin oxide or oxynitride film 1103 is deposited, as illustrated in FIG. 11, by a conventional CVD process. Typically oxide or oxynitride film 1103 is deposited at a thickness of about 30 Å to about 60 Å.

[0049] Subsequently, a mask 1201, e.g., photoresist or hard mask is applied over the PMOS transistor, as illustrated in FIG. 12, and the oxide or oxynitride film 1103 and high compressive stress silicon nitride film 1101 are removed from the NMOS transistor.

[0050] Adverting to FIG. 13, the mask 1201 is removed from the PMOS transistor. A highly tensile stressed silicon nitride film 1301, having a tensile stress of greater than 1.5 GPa, is then deposited over both the PMOS and NMOS transistors. Deposition of highly tensile stressed film 1301 can be implemented in a conventional manner. A thin oxide or oxynitride film 1303 is then deposited by a conventional CVD process, as at a thickness of about 30 Å to about 60 Å.

[0051] Adverting to FIG. 14, a mask 1401, such as a photoresist or hard mask, is then applied over the NMOS transistor, and the oxide or oxynitride film 1303 and the highly tensile stressed silicon nitride film 1301 are selectively removed from the PMOS transistor stopping on the oxide or oxynitride film 1103. The mask 1401 is then removed, and the resulting structure is illustrated in FIG. 15 which comprises an oxide or oxynitride film 1303 and a highly tensile stressed silicon nitride film 1301 over the NMOS transistor and an oxide or oxynitride film 1103 and a highly compressive stressed silicon nitride film 1101 over the PMOS transistor. The resulting CMOS device comprises both PMOS and NMOS transistors with increased channel carrier mobility and, hence, increased drive current, without increasing the resistivity of the transistors.

[0001] The embodiments of the present disclosure can achieve several technical effects, including resistance reduction, improved transistor and circuit

performance, and reduced power requirements. The present disclosure enjoys industrial applicability in any of various types of highly integrated semiconductor devices.

[0002] In the preceding description, the present disclosure is described with reference to specifically exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the present disclosure, as set forth in the claims. The specification and drawings are, accordingly, to be regarded as illustrative and not as restrictive. It is understood that the present disclosure is capable of using various other combinations and embodiments and is capable of any changes or modifications within the scope of the inventive concept as expressed herein.

WHAT IS CLAIMED IS:

1. A method comprising:
forming a transistor on a substrate, the transistor comprising source/drain regions and a gate electrode;
forming a first layer of nickel, containing a first amount of platinum, over the transistor;
forming a second layer of nickel, containing a second amount of platinum, on the first layer of nickel, wherein the second amount of platinum is greater than the first amount of platinum; and
annealing the first and second layers of nickel to form a layer of nickel silicide over the source/drain regions and the gate electrode, wherein the nickel silicide layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.
2. The method according to claim 1, comprising forming the first and second layers of nickel at a combined thickness of about from 45 Å to about 100 Å.
3. The method according to claim 2, comprising forming the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness.
4. The method according to claim 3, comprising annealing to form the layer of nickel silicide at a thickness of about 100 Å to about 220 Å.
5. The method according to claim 1, comprising annealing by heating at a temperature of about 340° C to about 450° C.
6. The method according to claim 5, comprising annealing by heating at a temperature of about 360° C to about 400° C.
7. The method according to claim 1, comprising annealing by:
 - (a) heating at a first temperature of about 280° C to about 350° C;

(b) followed by heating at a second temperature of about 360° C to about 450° C.

8. The method according to claim 7, comprising heating at the first temperature of about 300° C to about 320° C and heating at the second temperature of about 380° C to about 420° C.

9. The method according to claim 1, comprising forming the first layer of nickel with a platinum content of about 8 wt. % to about 12 wt. %.

10. The method according to claim 9, comprising forming the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %.

11. The method according to claim 10, comprising:
forming the first nickel layer of nickel at a platinum content of about 10 wt. %; and
forming the second layer of nickel at a platinum content of about 30 wt. % to about 50 wt. %.

12. The method according to claim 1, further comprising forming a capping layer of titanium nitride over the second layer of nickel before annealing.

13. The method according to claim 1, further comprising:
removing the capping layer after annealing; and
etching to remove unreacted nickel.

14. A semiconductor device comprising:
a silicon substrate,
a transistor, comprising source/drain regions and a gate electrode, on the substrate;
and
a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

15. The semiconductor device according to claim 14, further comprising:

a P-channel transistor;

an N-channel transistor;

a first nickel silicide layer on the P-channel transistor;

a second nickel silicide layer on the N-channel transistor;

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient with platinum increasing in amount in a direction away from the P-channel and N-channel transistors;

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

16. The semiconductor device according to claim 14, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

17. The semiconductor device according to claim 16, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

18. The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

19. The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

20. A method comprising:

forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate electrode and source/drain regions formed in the silicon substrate;

depositing a first layer of nickel containing a first amount of platinum on both transistors;

depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, wherein the second amount of platinum is greater than the first amount of platinum;

depositing a titanium nitride capping layer on the second layer of nickel;

annealing the first and second layers of nickel to form a layer of nickel silicide containing platinum having a concentration gradient with platinum increasing in amount in a direction away from the gate electrode and source/drain regions,

etching to remove the capping layer and unreacted nickel;

forming a first stress liner exhibiting high compressive stress over both transistors;

selectively removing the first stress liner from the N-channel transistor;

forming a second stress liner exhibiting high tensile stress over both transistors; and

selectively removing the second stress liner from the P-channel transistor.

ABSTRACT OF DISCLOSURE

Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

COMBINED DECLARATION AND POWER OF ATTORNEY
FOR PATENT APPLICATION

As a below named inventor, I hereby declare that:

My residence, post office address and citizenship are as stated below, next to my name.

I believe I am the original, first, and joint inventor of the subject matter which is claimed and for which a patent is sought on the invention entitled:

METHOD TO REDUCE MOL DAMAGE ON NISI

the specification of which is attached hereto.

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claim(s), as amended by any amendment referred to above.

I acknowledge the duty to disclose all information known to me to be material to patentability as defined in Title 37, Code of Federal Regulations, Section 1.56 (copy attached).

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d), on any foreign application(s) for patent or inventor's certificate listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s)		Priority Claimed		
(Number)	(Country)	(Day/Month/Year Filed)	Yes	No

I hereby claim the benefit under Title 35, United States Code, Section 119(e) of any United States provisional application(s) listed below:

I hereby claim benefit under Title 35, United States Code, Section 120 of any United States application(s) listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose all information known to me to be material to patentability as defined in Title 37, Code of Federal Regulations, Section 1.56 (copy attached) which became available between the filing date of the prior application and the national or PCT International filing date of this application:

As named inventor, I hereby revoke all powers of attorney previously given and appoint the practitioners associated with the following Customer No.:

Customer Number:

30671

with full power of substitution and revocation, to prosecute this application and to transact all business in the Patent and Trademark Office connected therewith.

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

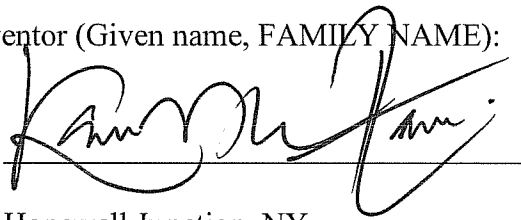
Send all future correspondence to the attention of Arthur J. Steiner, Reg. No. 26106, care of the address associated with the following customer number, and direct all telephone calls to the same at (703) 519-9954.

Customer Number:

30671

Full Name of First Inventor (Given name, FAMILY NAME): Karthik RAMANI

Inventor's Signature



Date

29 Jan 2009

Residence:

Hopewell Junction, NY

Citizenship India

Post Office Address 3603 Chelsea Cove South, Hopewell Junction, NY 12533

Full Name of Second Inventor (given name, FAMILY NAME): Paul R. BESSER

Inventor's Signature

Date

Residence:

Sunnyvale, CA

Citizenship US

Post Office Address 1087 Yorktown Drive, Sunnyvale, CA 94087

Title 37, Code of Federal Regulations, Section 1.56
Duty to Disclose Information Material to Patentability

(a) A patent by its very nature is affected with a public interest. The public interest is best served, and the most effective patent examination occurs when, at the time an application is being examined, the Office is aware of and evaluates the teachings of all information material to patentability. Each individual associated with the filing and prosecution of a patent application has a duty of candor and good faith in dealing with the Office, which includes a duty to disclose to the Office all information known to that individual to be material to patentability as defined in this section. The duty to disclose information exists with respect to each pending claim until the claim is canceled or withdrawn from consideration, or the application becomes abandoned. Information material to the patentability of a claim that is canceled or withdrawn from consideration need not be submitted if the information is not material to the patentability of any claim remaining under consideration in the application. There is no duty to submit information which is not material to the patentability of any existing claim. The duty to disclose all information known to be material to patentability is deemed to be satisfied if all information known to be material to patentability of any claim issued in a patent was cited by the Office or submitted to the Office in the manner prescribed by §§ 1.97(b)-(d) and 1.98. However, no patent will be granted on an application in connection with which fraud on the Office was practiced or attempted or the duty of disclosure was violated through bad faith or intentional misconduct. The Office encourages applicants to carefully examine:

- (1) Prior art cited in search reports of a foreign patent office in a counterpart application, and
- (2) The closest information over which individuals associated with the filing or prosecution of a patent application believe any pending claim patentably defines, to make sure that any material information contained therein is disclosed to the Office.

(b) Under this section, information is material to patentability when it is not cumulative to information already of record or being made of record in the application, and

- (1) It establishes, by itself or in combination with other information, a prima facie case of unpatentability of a claim; or
- (2) It refutes, or is inconsistent with, a position the applicant takes in:
 - (i) Opposing an argument of unpatentability relied on by the Office, or
 - (ii) Asserting an argument of patentability.

A prima facie case of unpatentability is established when the information compels a conclusion that a claim is unpatentable under the preponderance of evidence, burden-of-proof standard, giving each term in the claim its broadest reasonable construction consistent with the specification, and before any consideration is given to evidence which may be submitted in an attempt to establish a contrary conclusion of patentability.

(c) Individuals associated with the filing or prosecution of a patent application within the meaning of this section are:

- (1) Each inventor named in the application;
- (2) Each attorney or agent who prepares or prosecutes the application; and
- (3) Every other person who is substantively involved in the preparation or prosecution of the application and who is associated with the inventor, with the assignee or with anyone to whom there is an obligation to assign the application.

(d) Individuals other than the attorney, agent or inventor may comply with this section by disclosing information to the attorney, agent, or inventor.

COMBINED DECLARATION AND POWER OF ATTORNEY
FOR PATENT APPLICATION

As a below named inventor, I hereby declare that:

My residence, post office address and citizenship are as stated below, next to my name.

I believe I am the original, first, and joint inventor of the subject matter which is claimed and for which a patent is sought on the invention entitled:

METHOD TO REDUCE MOL DAMAGE ON NISI

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I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claim(s), as amended by any amendment referred to above.

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Prior Foreign Application(s)		Priority Claimed		
_____ (Number)	_____ (Country)	_____ (Day/Month/Year Filed)	_____ Yes	_____ No

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As named inventor, I hereby revoke all powers of attorney previously given and appoint the practitioners associated with the following Customer No.:

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I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

Send all future correspondence to the attention of Arthur J. Steiner, Reg. No. 26106, care of the address associated with the following customer number, and direct all telephone calls to the same at (703) 519-9954.

Customer Number:

30671

Full Name of First Inventor (Given name, FAMILY NAME): Karthik RAMANI

Inventor's Signature _____ Date _____

Residence: Hopewell Junction, NY Citizenship India

Post Office Address 3603 Chelsea Cove South, Hopewell Junction, NY 12533

Full Name of Second Inventor (given name, FAMILY NAME): Paul R. BESSER

Inventor's Signature Paul R. Besser Date 1/29/09

Residence: Sunnyvale, CA Citizenship US

Post Office Address 1087 Yorktown Drive, Sunnyvale, CA 94087

Title 37, Code of Federal Regulations, Section 1.56
Duty to Disclose Information Material to Patentability

(a) A patent by its very nature is affected with a public interest. The public interest is best served, and the most effective patent examination occurs when, at the time an application is being examined, the Office is aware of and evaluates the teachings of all information material to patentability. Each individual associated with the filing and prosecution of a patent application has a duty of candor and good faith in dealing with the Office, which includes a duty to disclose to the Office all information known to that individual to be material to patentability as defined in this section. The duty to disclose information exists with respect to each pending claim until the claim is canceled or withdrawn from consideration, or the application becomes abandoned. Information material to the patentability of a claim that is canceled or withdrawn from consideration need not be submitted if the information is not material to the patentability of any claim remaining under consideration in the application. There is no duty to submit information which is not material to the patentability of any existing claim. The duty to disclose all information known to be material to patentability is deemed to be satisfied if all information known to be material to patentability of any claim issued in a patent was cited by the Office or submitted to the Office in the manner prescribed by §§ 1.97(b)-(d) and 1.98. However, no patent will be granted on an application in connection with which fraud on the Office was practiced or attempted or the duty of disclosure was violated through bad faith or intentional misconduct. The Office encourages applicants to carefully examine:

- (1) Prior art cited in search reports of a foreign patent office in a counterpart application, and
- (2) The closest information over which individuals associated with the filing or prosecution of a patent application believe any pending claim patentably defines, to make sure that any material information contained therein is disclosed to the Office.

(b) Under this section, information is material to patentability when it is not cumulative to information already of record or being made of record in the application, and

- (1) It establishes, by itself or in combination with other information, a prima facie case of unpatentability of a claim; or
- (2) It refutes, or is inconsistent with, a position the applicant takes in:
 - (i) Opposing an argument of unpatentability relied on by the Office, or
 - (ii) Asserting an argument of patentability.

A prima facie case of unpatentability is established when the information compels a conclusion that a claim is unpatentable under the preponderance of evidence, burden-of-proof standard, giving each term in the claim its broadest reasonable construction consistent with the specification, and before any consideration is given to evidence which may be submitted in an attempt to establish a contrary conclusion of patentability.

(c) Individuals associated with the filing or prosecution of a patent application within the meaning of this section are:

- (1) Each inventor named in the application;
- (2) Each attorney or agent who prepares or prosecutes the application; and
- (3) Every other person who is substantively involved in the preparation or prosecution of the application and who is associated with the inventor, with the assignee or with anyone to whom there is an obligation to assign the application.

(d) Individuals other than the attorney, agent or inventor may comply with this section by disclosing information to the attorney, agent, or inventor.

Electronic Acknowledgement Receipt

EFS ID:	4738862
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	30671
Filer:	Arthur J. Steiner/Dara Phommachanh
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	NY0281 (01158-1030)
Receipt Date:	05-FEB-2009
Filing Date:	
Time Stamp:	17:04:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1090
RAM confirmation Number	2617
Deposit Account	504213
Authorized User	DITTHAVONG,PHOUPHANOMKETH

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NY0281_01158-1030_ADS.pdf	17243 fc5465844603ce578688cc0be2cf0c27d7e8e503	no	2
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2		NY0281_01158-1030_Specifica tion.pdf	101083 1b3ca6a68e8f9b681d65651c00625131096152b5	yes	20
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	15	
	Claims		16	19	
	Abstract		20	20	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	NY0281_01158-1030_Drawings .pdf	41594 381f400879c3fe704dd5df267f73309d0e13e953	no	13
Warnings:					
Information:					
4	Oath or Declaration filed	NY0281_01158-1030_Declarati on.pdf	414302 e2a5402509fce23157be284b75922e784e08e49f	no	6
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
5	Information Disclosure Statement Letter	NY0281_01158-1030_IDSTrans mittalLetter.pdf	23457 f5c400b100a4494d53c49ddb4f58a2059585e495	no	2
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Information:					
6	Information Disclosure Statement (IDS) Filed (SB/08)	NY0281_01158-1030_IDS_PTO _SB08.pdf	607940 0b633261f0b096c2cf48bab9d824a7c9d4aa373	no	4
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Information:					

7	Fee Worksheet (PTO-06)	fee-info.pdf	33051 538e0ed43922559182ec80712699e680d9e ebb47	no	2
Warnings:					
Information:					
Total Files Size (in bytes):				1238670	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

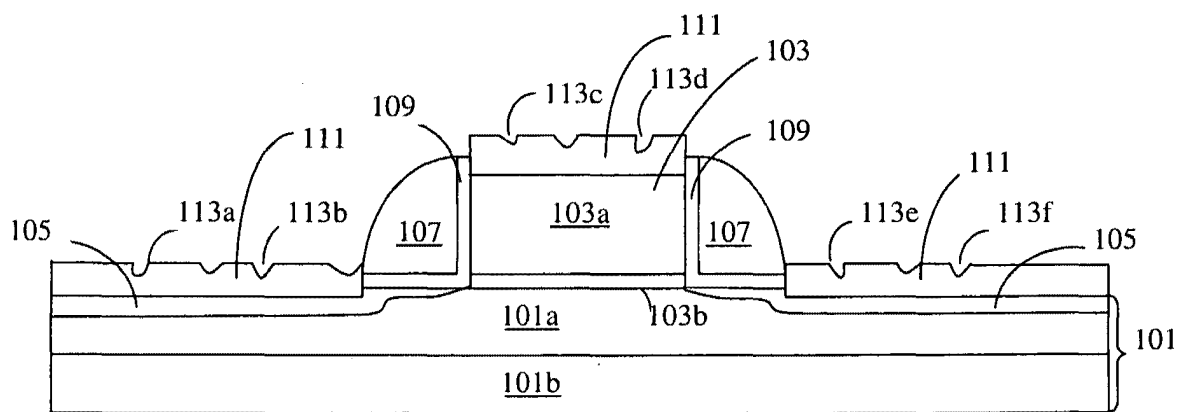


FIG. 1

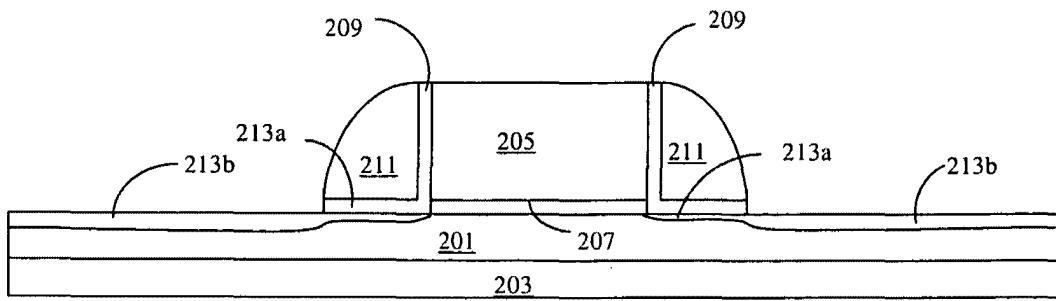


FIG. 2

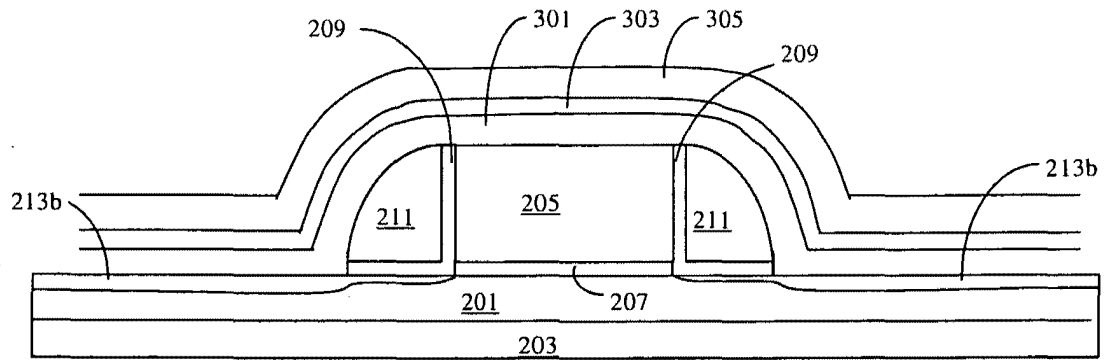


FIG. 3

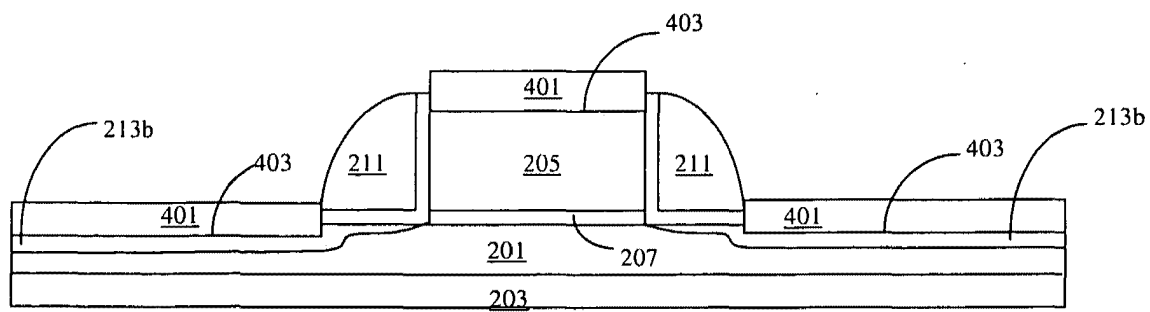


FIG. 4

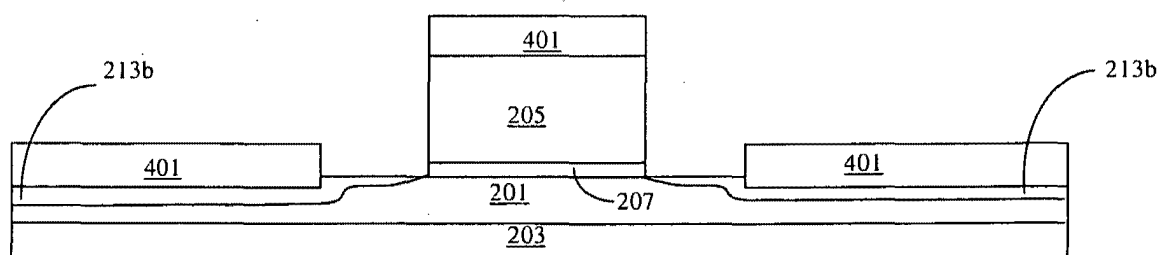


FIG. 5

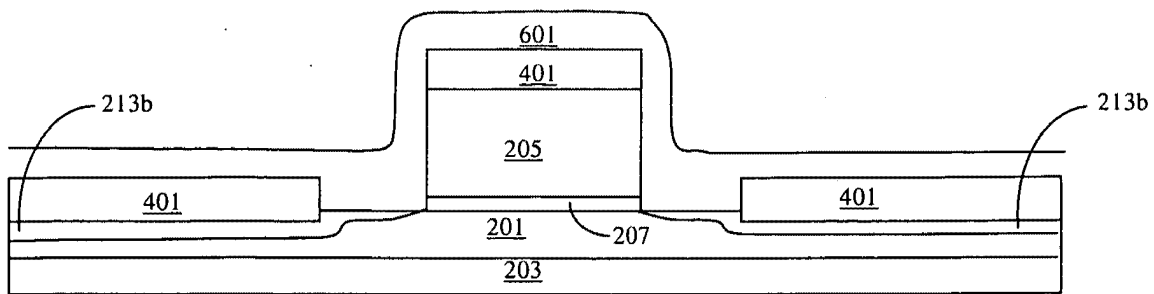


FIG. 6

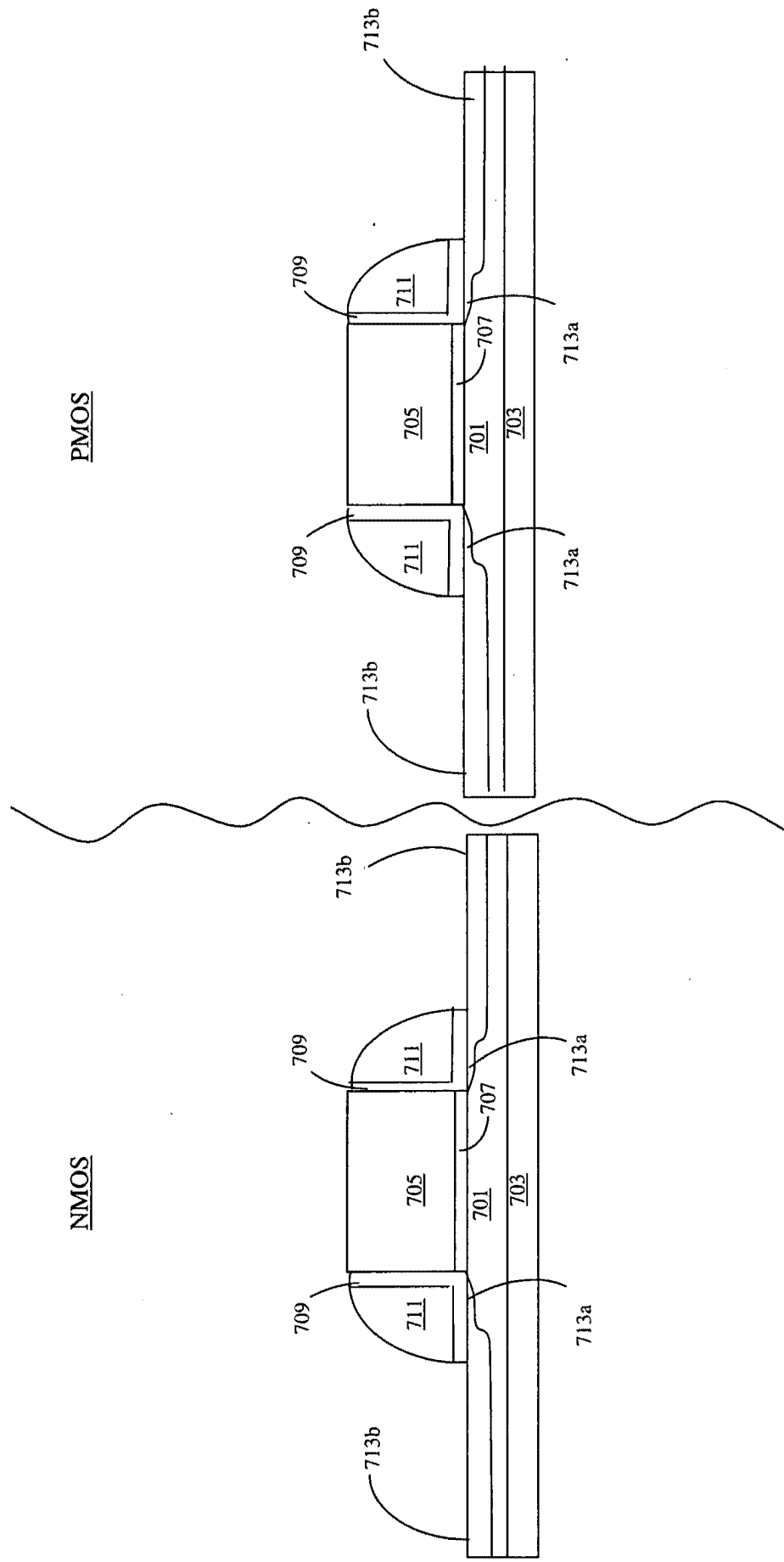


FIG. 7

NMOS

PMOS

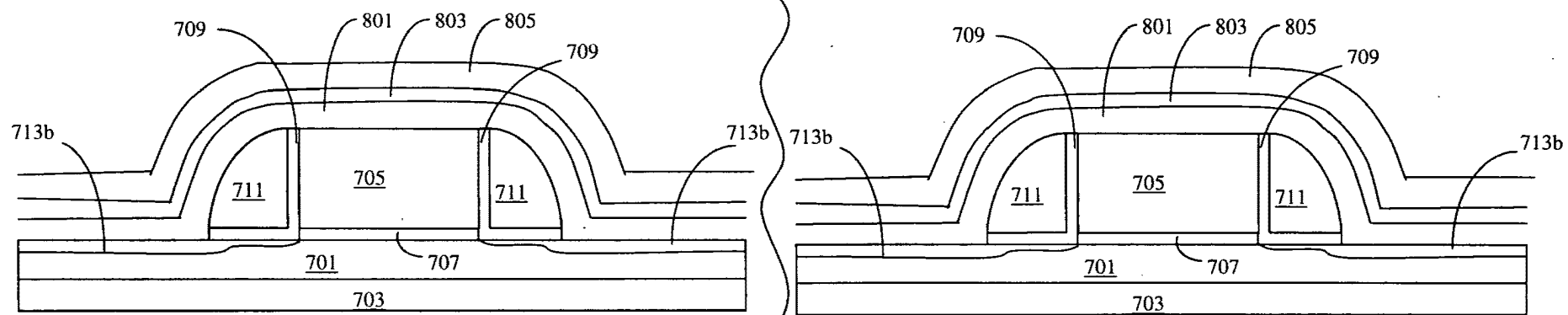


FIG. 8

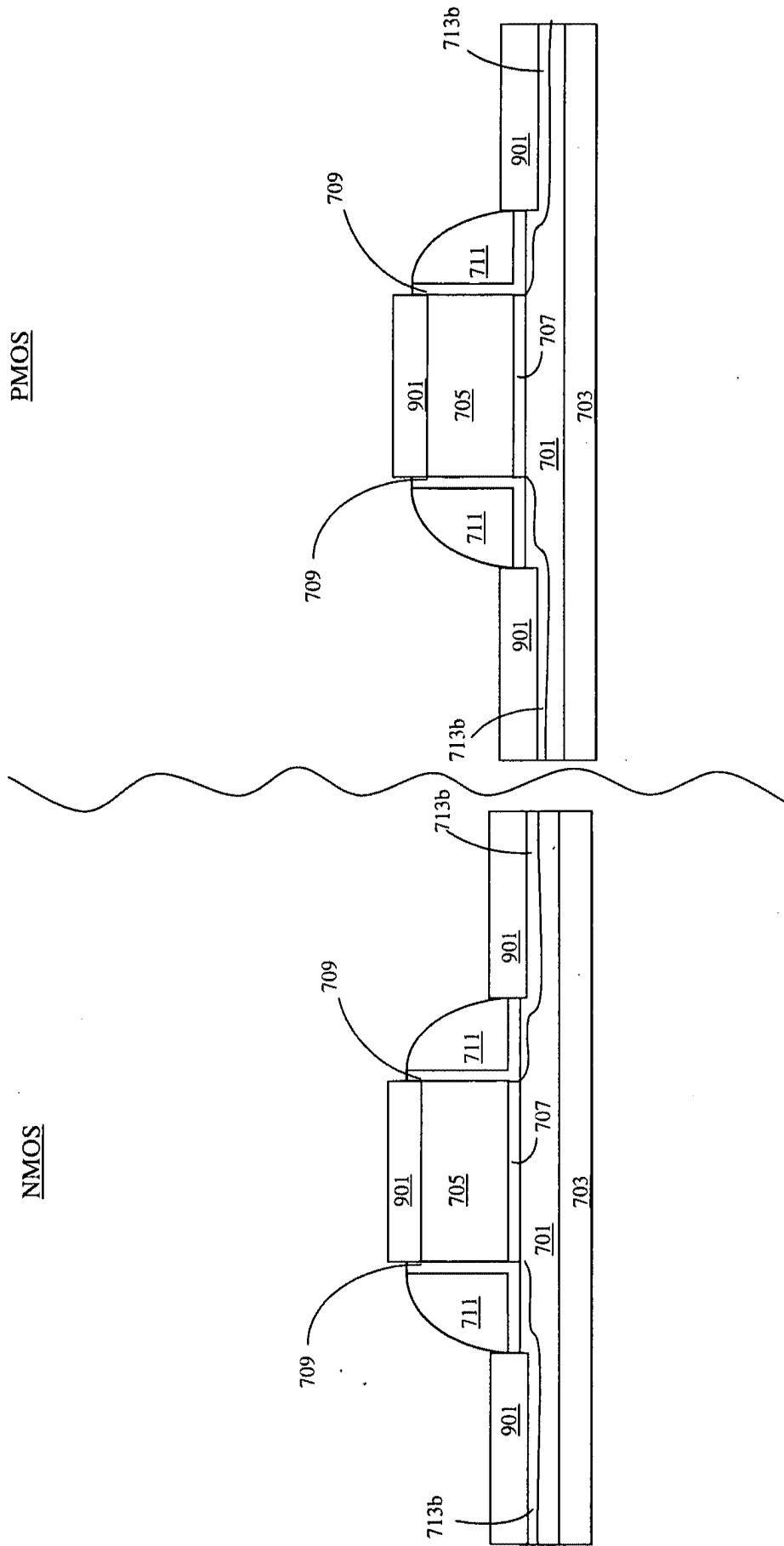


FIG. 9

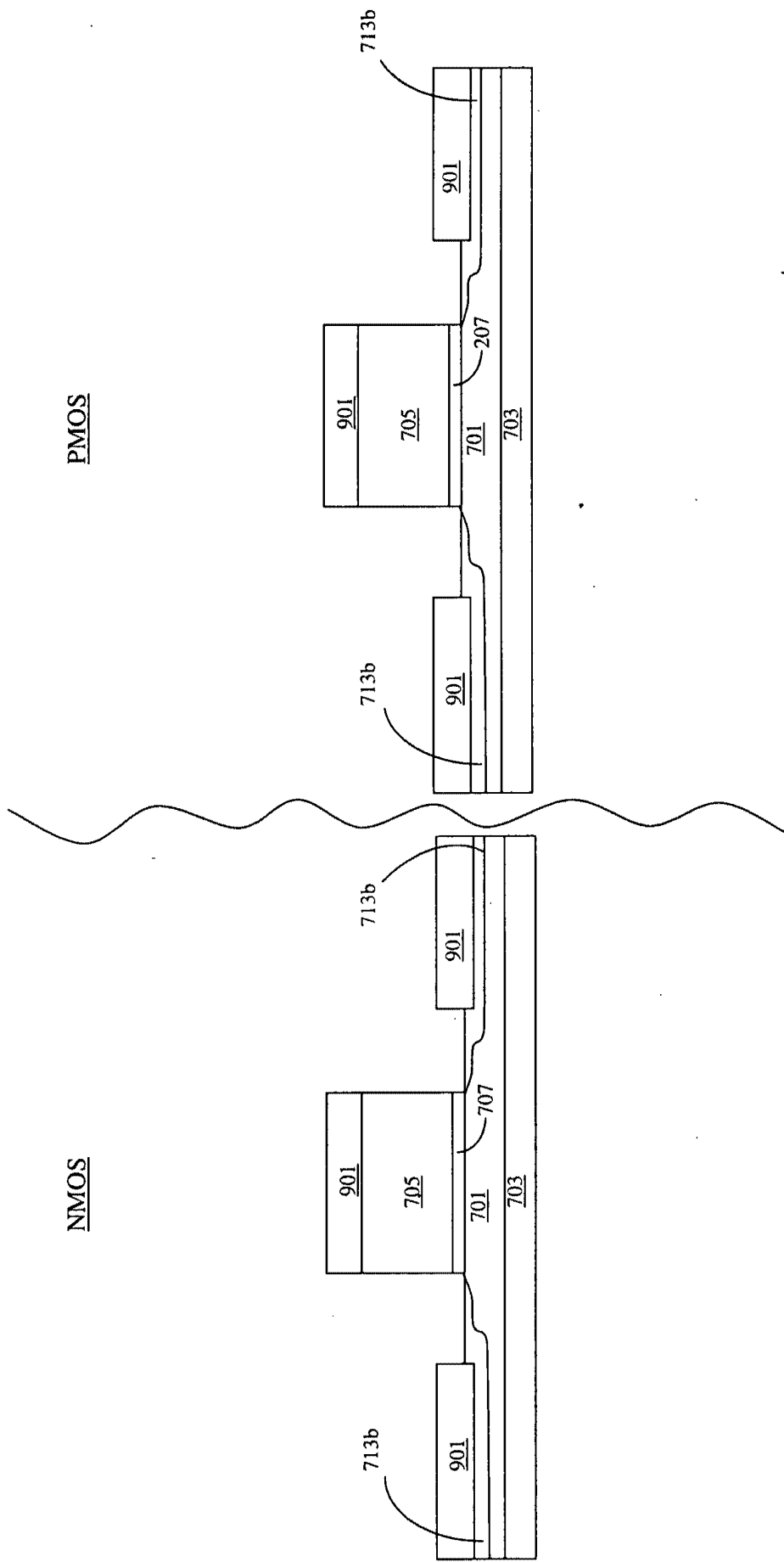


FIG. 10

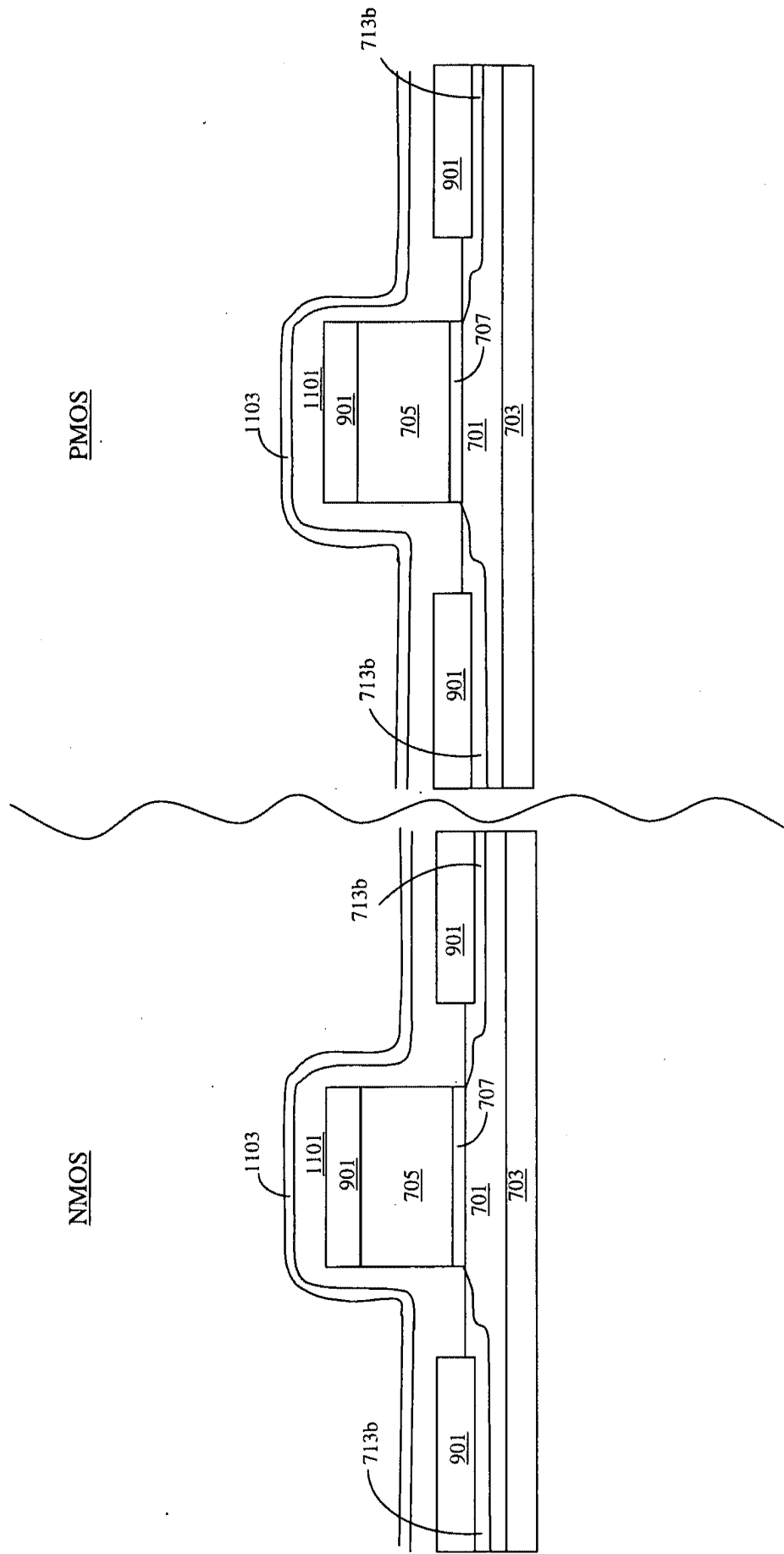


FIG. 11

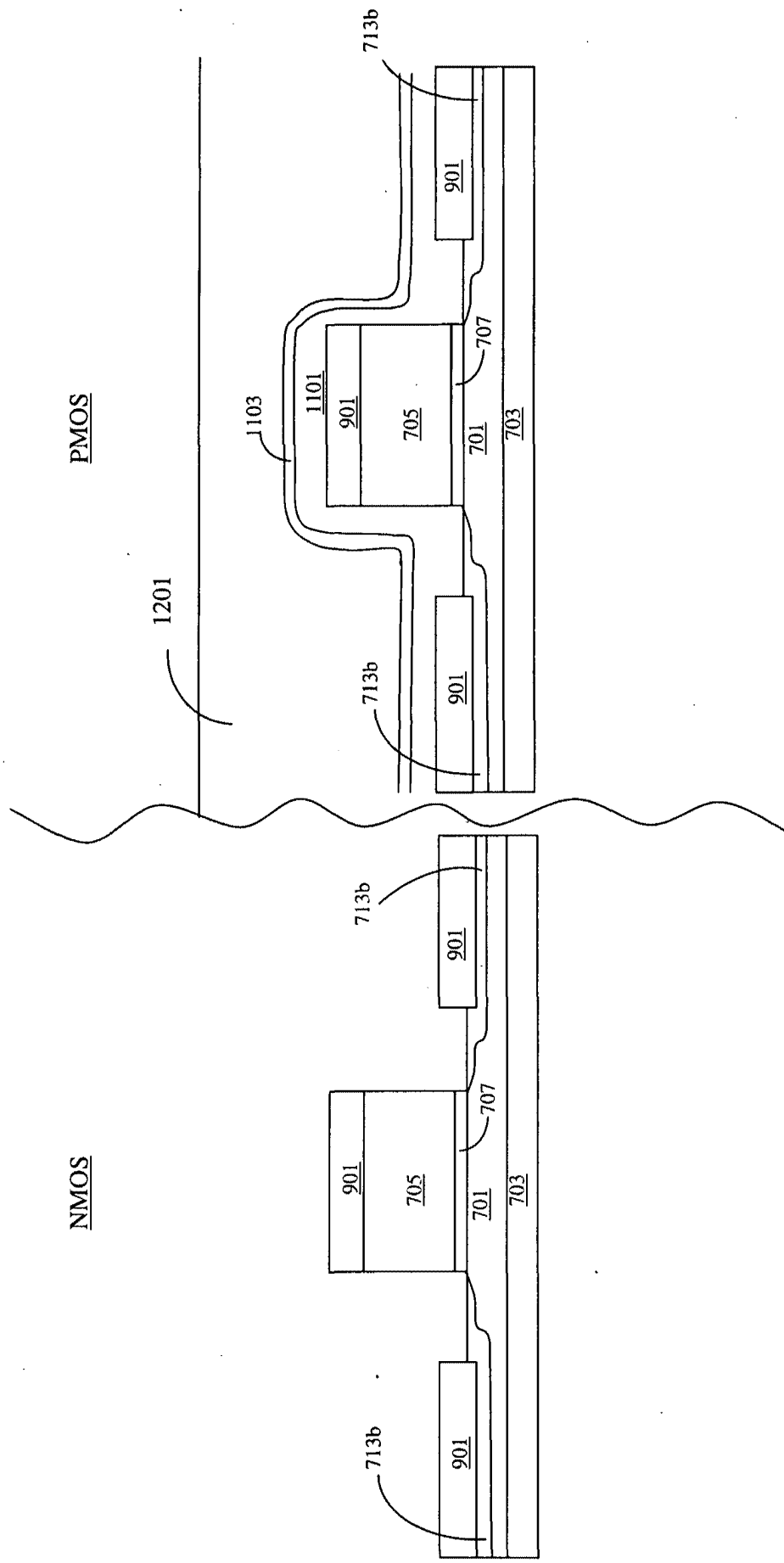


FIG. 12

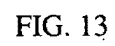


FIG. 13

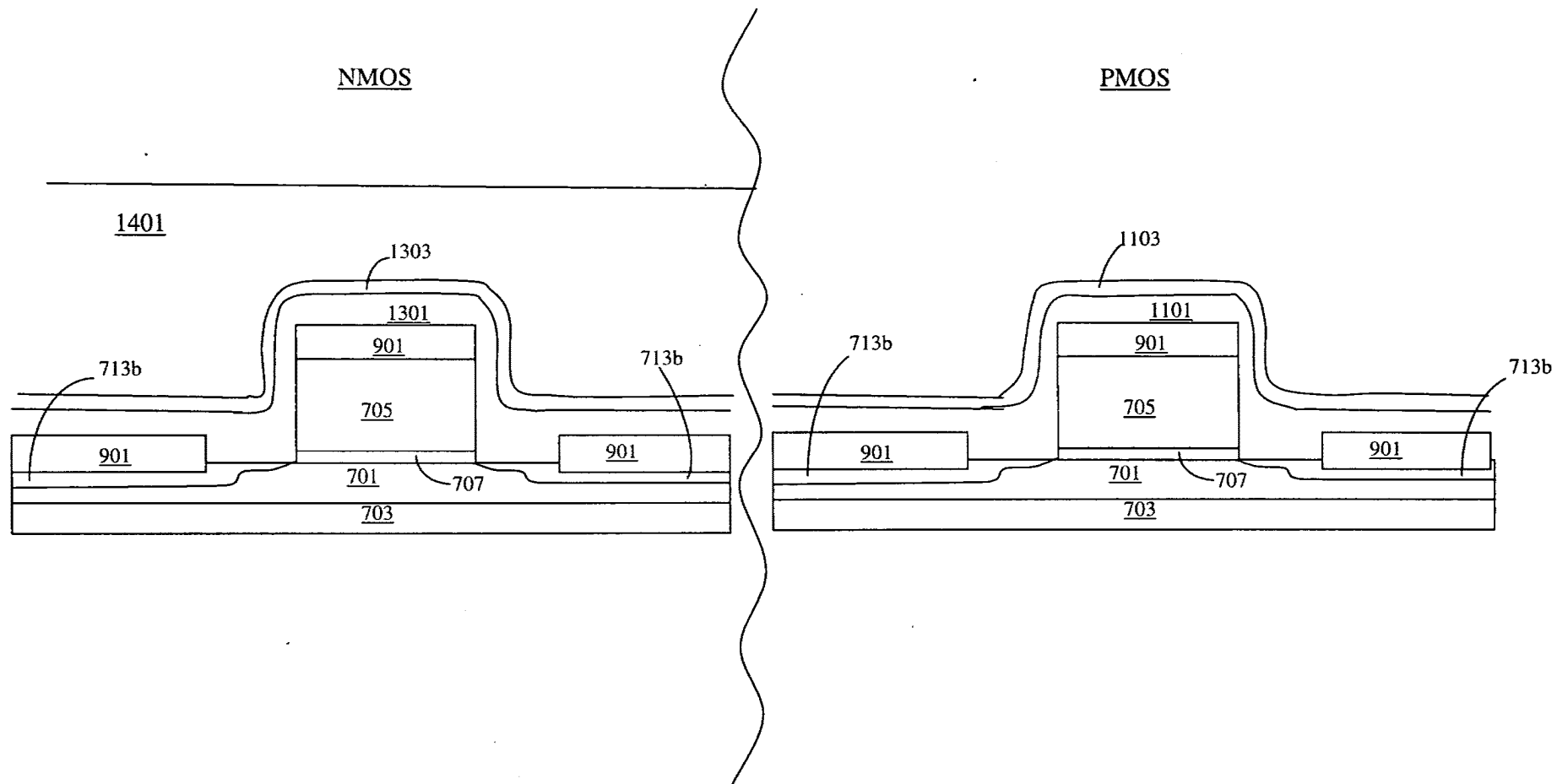


FIG. 14

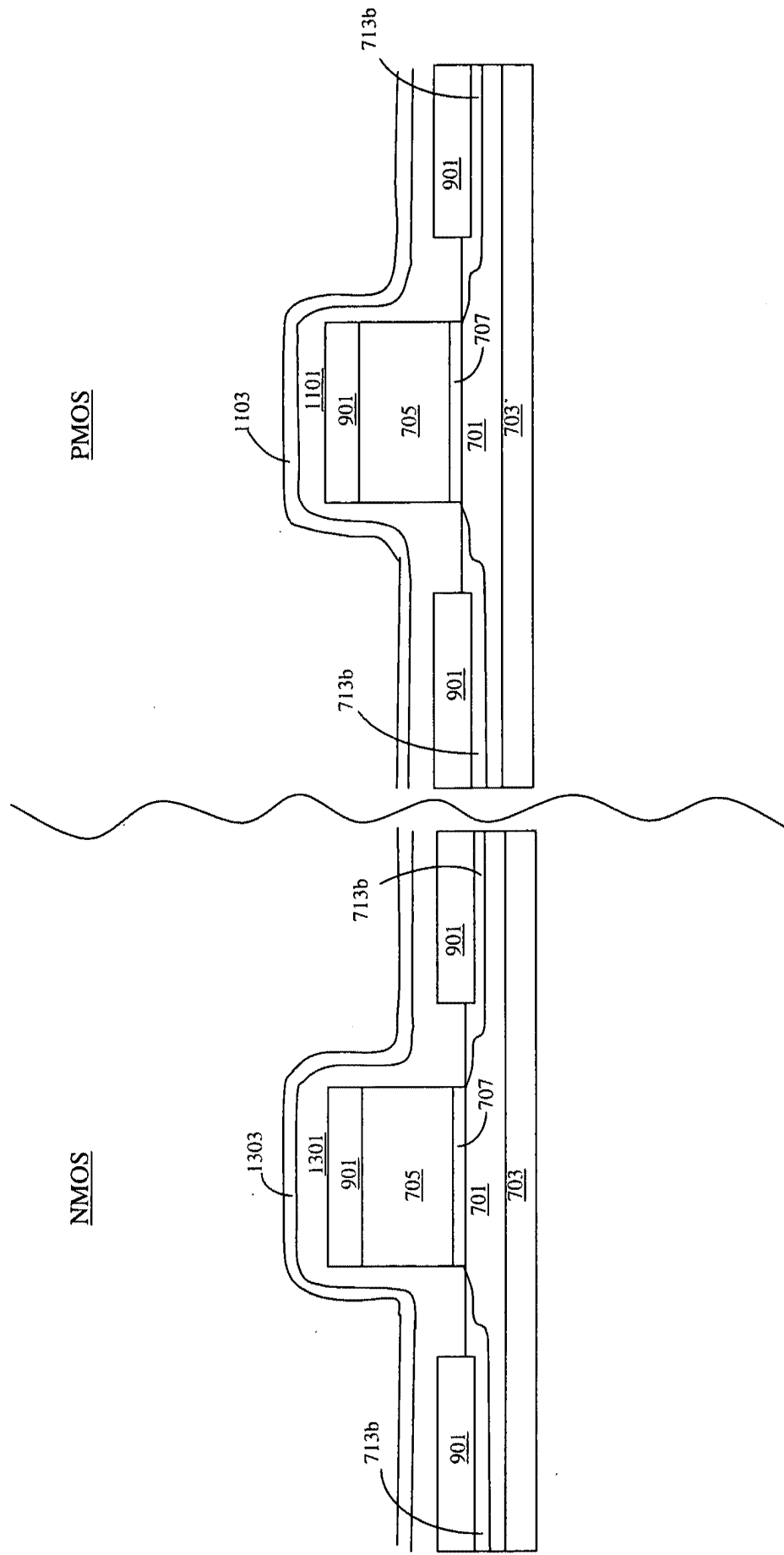


FIG. 15

Filing Date: 02/05/09

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/366,378	
APPLICATION AS FILED – PART I						
(Column 1)		(Column 2)		SMALL ENTITY		OR
OTHER THAN SMALL ENTITY						
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A		N/A	330
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A		N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A		N/A	220
TOTAL CLAIMS (37 CFR 1.16(i))	20	minus 20 =	x\$26		x\$52	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 = *	x\$110		x\$220	
APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR				
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				195		390
			TOTAL		TOTAL	1090
* If the difference in column 1 is less than zero, enter "0" in column 2.						
APPLICATION AS AMENDED – PART II						
(Column 1)		(Column 2)		(Column 3)		
SMALL ENTITY		OTHER THAN SMALL ENTITY				
RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)			
X =		X =				
X =		X =				
N/A		N/A				
TOTAL		TOTAL				
ADD'T FEE		ADD'T FEE				
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/366,378	02/05/2009	1794	1090	01158-1030	20	3

CONFIRMATION NO. 7721

30671
DITTHAVONG MORI & STEINER, P.C.
918 Prince St.
Alexandria, VA 22314

FILING RECEIPT



OC000000034652309

Date Mailed: 02/26/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Karthik Ramani, Hopewell Junction, NY;
Paul R. Besser, Sunnyvale, CA;

Assignment For Published Patent Application

Advanced Micro Devices, Inc., Sunnyvale, CA

Power of Attorney: The patent practitioners associated with Customer Number 30671

Domestic Priority data as claimed by applicant

Foreign Applications

If Required, Foreign Filing License Granted: 02/19/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/366,378**

Projected Publication Date: 08/05/2010

Non-Publication Request: No

Early Publication Request: No

Title

METHOD TO REDUCE MOL DAMAGE ON NiSi

Preliminary Class

428

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	7721
Application Ser. No.:	12/366,378	Examiner:	Not yet assigned
Filed:	February 5, 2009	Group Art Unit:	1794

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Dear Sir:

Prior to consideration, please amend the above-identified application based on the following:

AMENDMENT TO THE DRAWINGS 2

REMARKS 3

AMENDMENT TO THE DRAWINGS

The attached Replacement Sheets of drawings include FIGS. 1-15. These Replacement Sheets, which include FIGS 1-15, replace the original sheet including FIGS. 1-15.

Attachment: Replacement Sheets (13)

REMARKS

Favorable consideration of this application in light of the following discussion is respectfully requested.

Replacement Sheets for FIGS. 1-15 have been submitted herewith. No new matter is introduced herein.

The Applicants are submitting herewith Replacement Sheets for FIGS. 1-15 that clearly satisfy the requirements of 37 CFR §§1.84 and 1.121(d). The Replacements Sheets include formal versions of the originally submitted drawings that meet all of the above requirements.

Favorable consideration is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9957 so that such issues may be resolved as expeditiously as possible.

To the extent necessary, a petition for an extension of time under 37 CFR §1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,
DITTHAVONG MORI & STEINER, P.C.

March 11, 2009
Date

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106

918 Prince Street
Alexandria, VA 22314
Tel. (703) 519-9957
Fax. (703) 519-9958

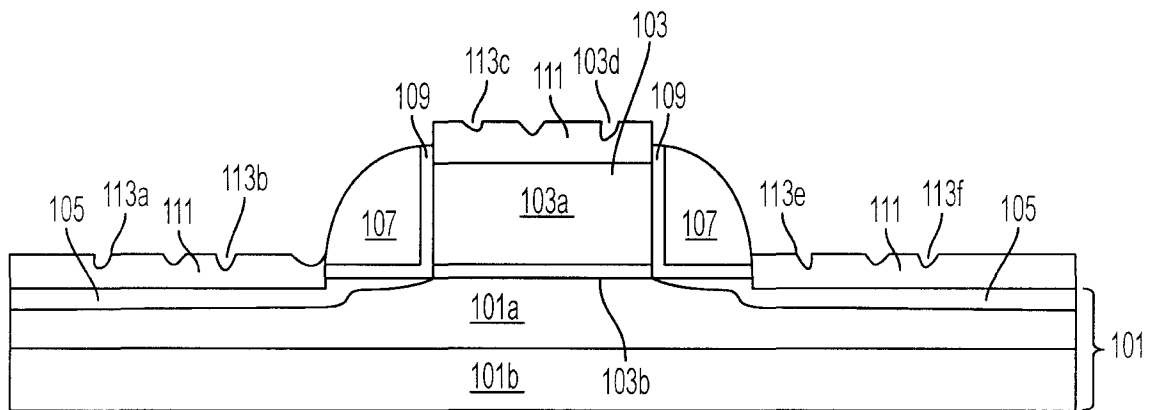


FIG. 1

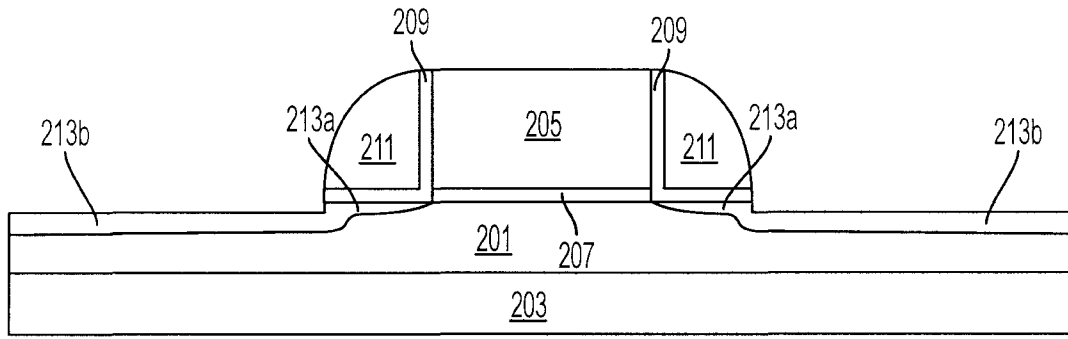


FIG. 2

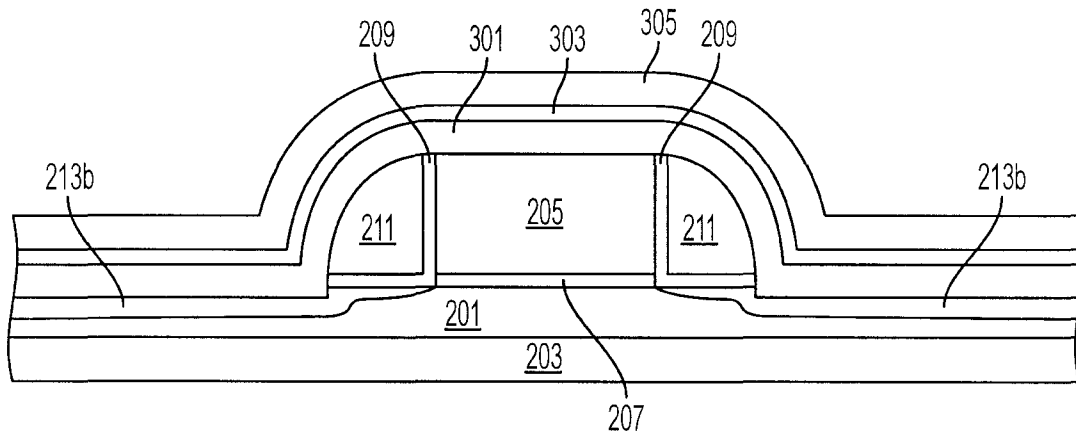


FIG. 3

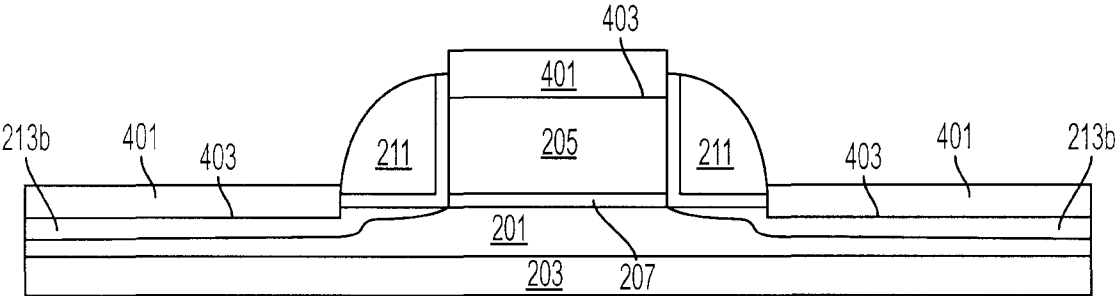


FIG. 4

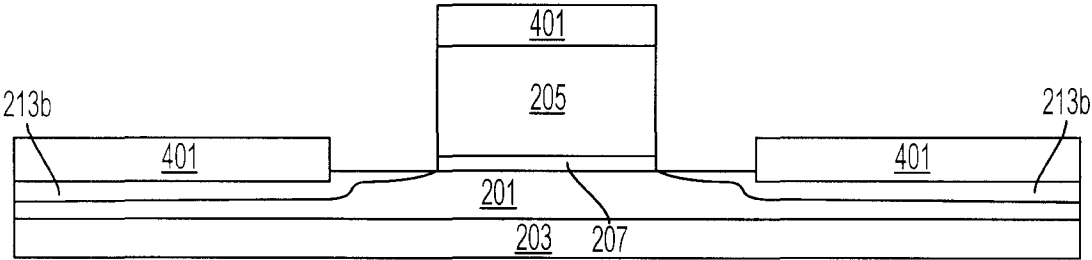


FIG. 5

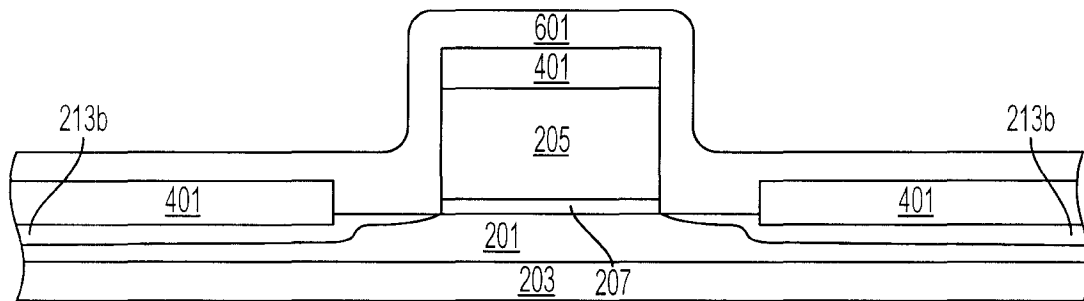


FIG. 6

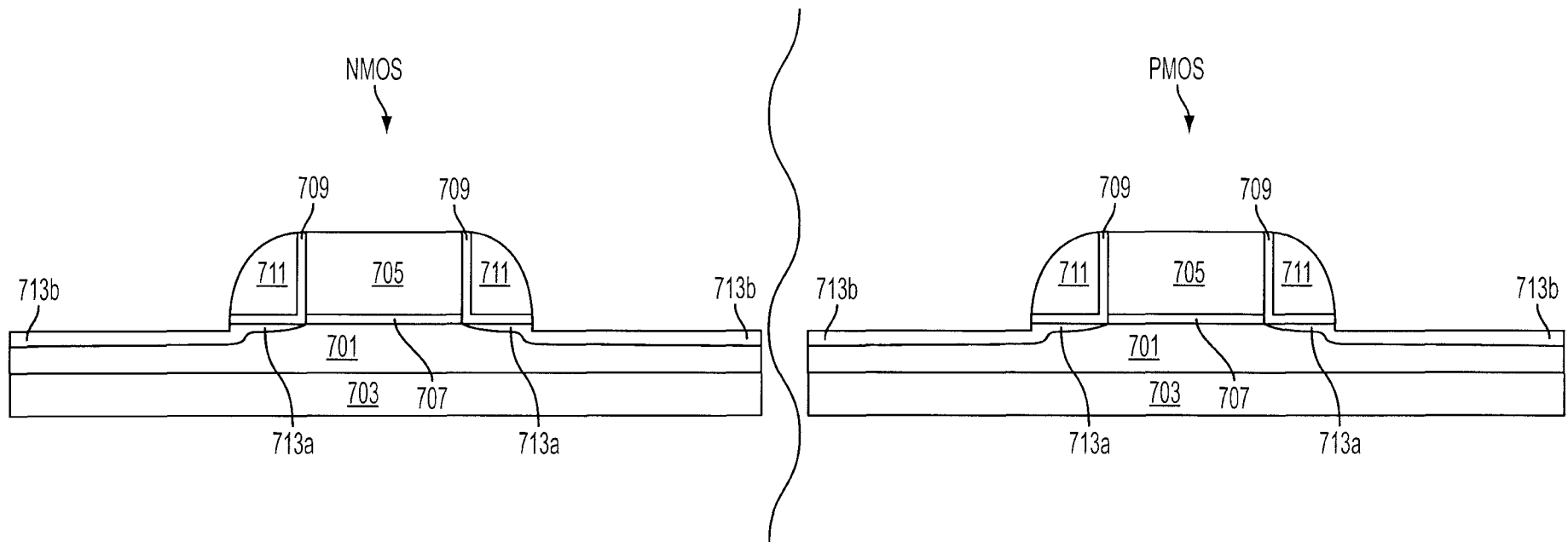


FIG. 7

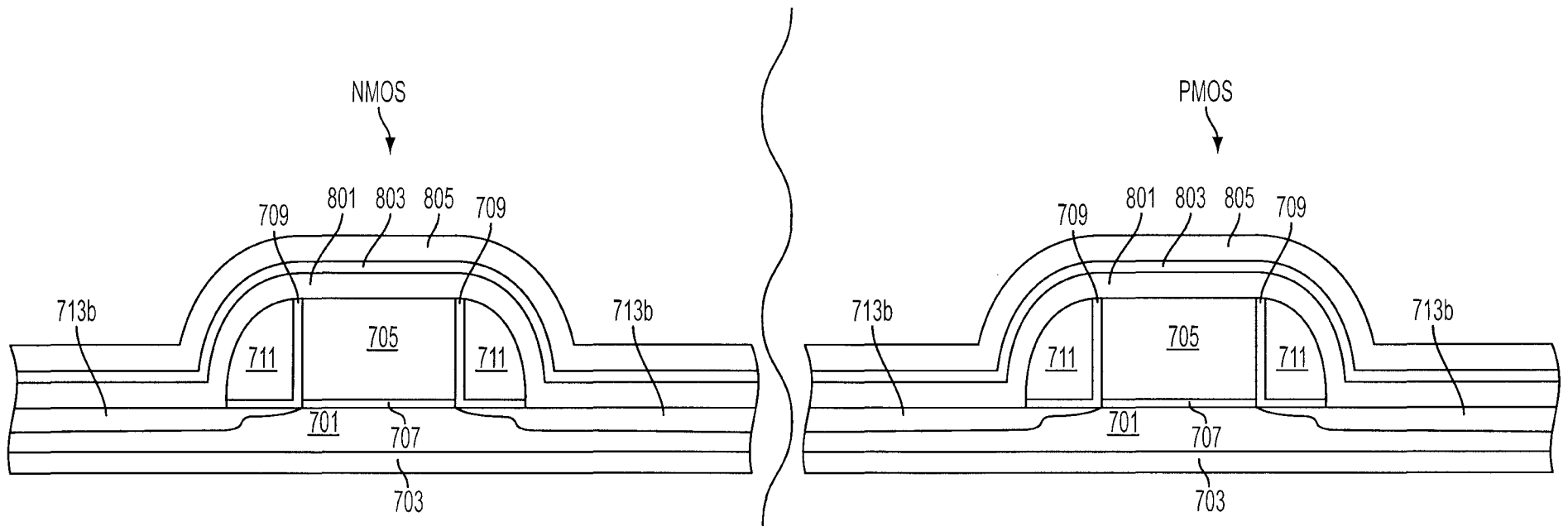


FIG. 8

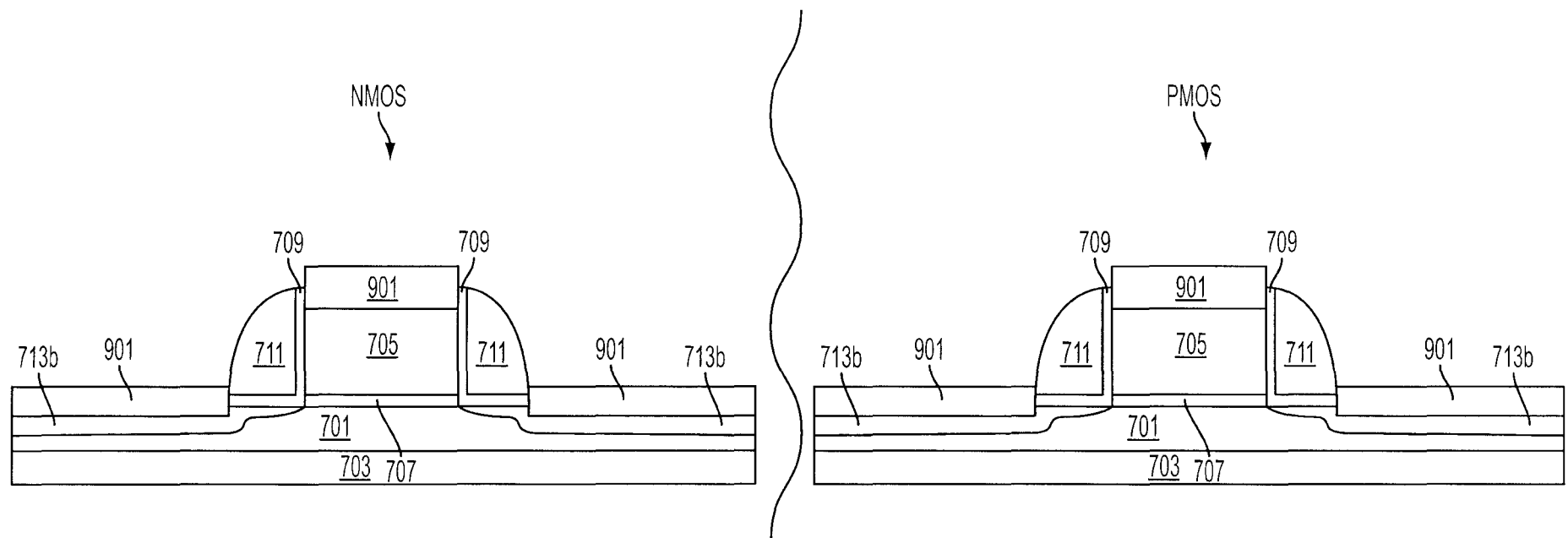


FIG. 9

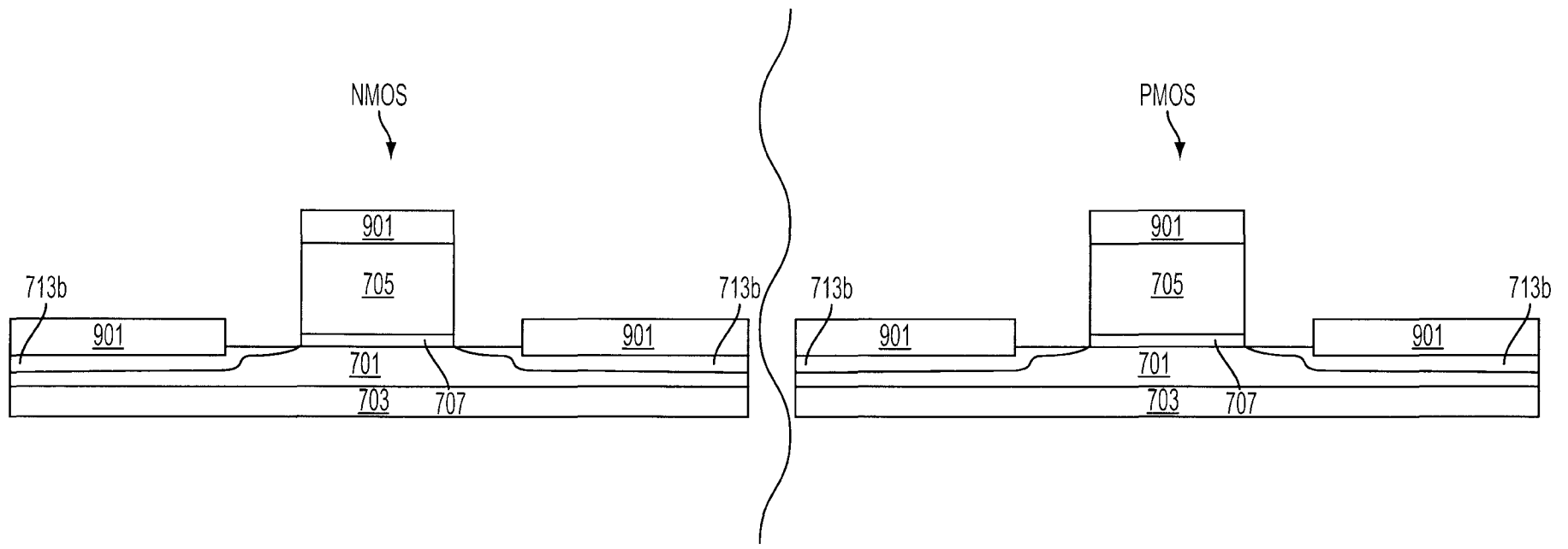


FIG. 10

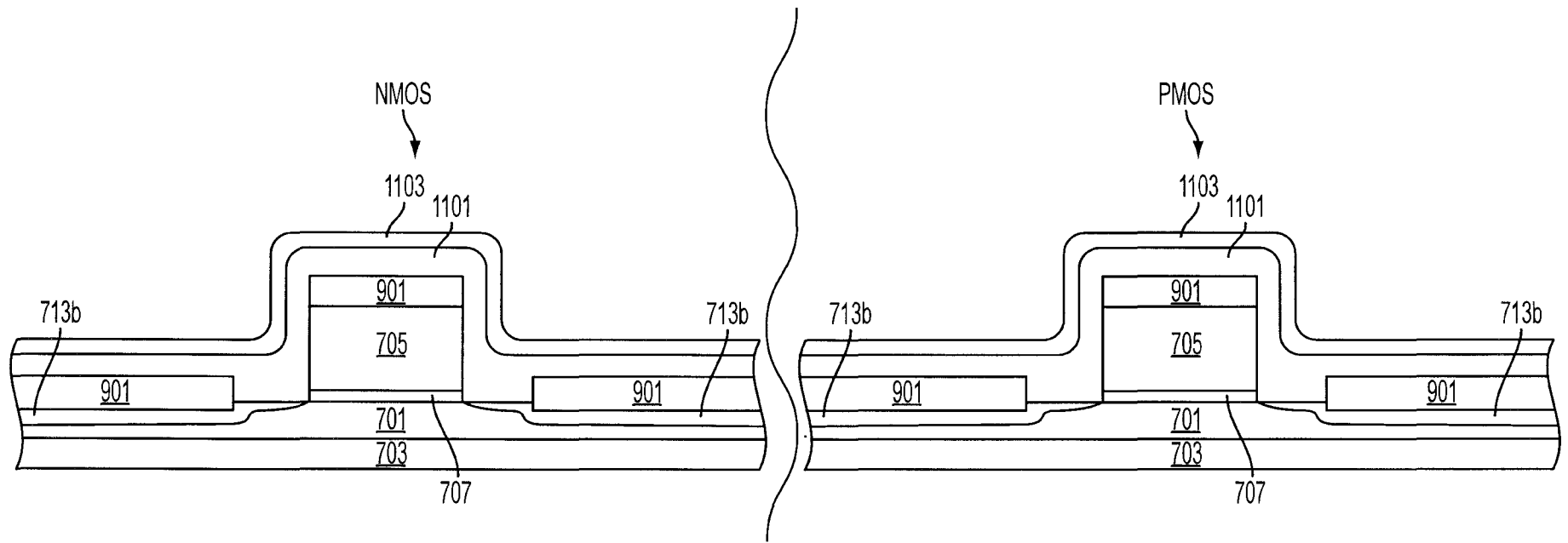


FIG. 11

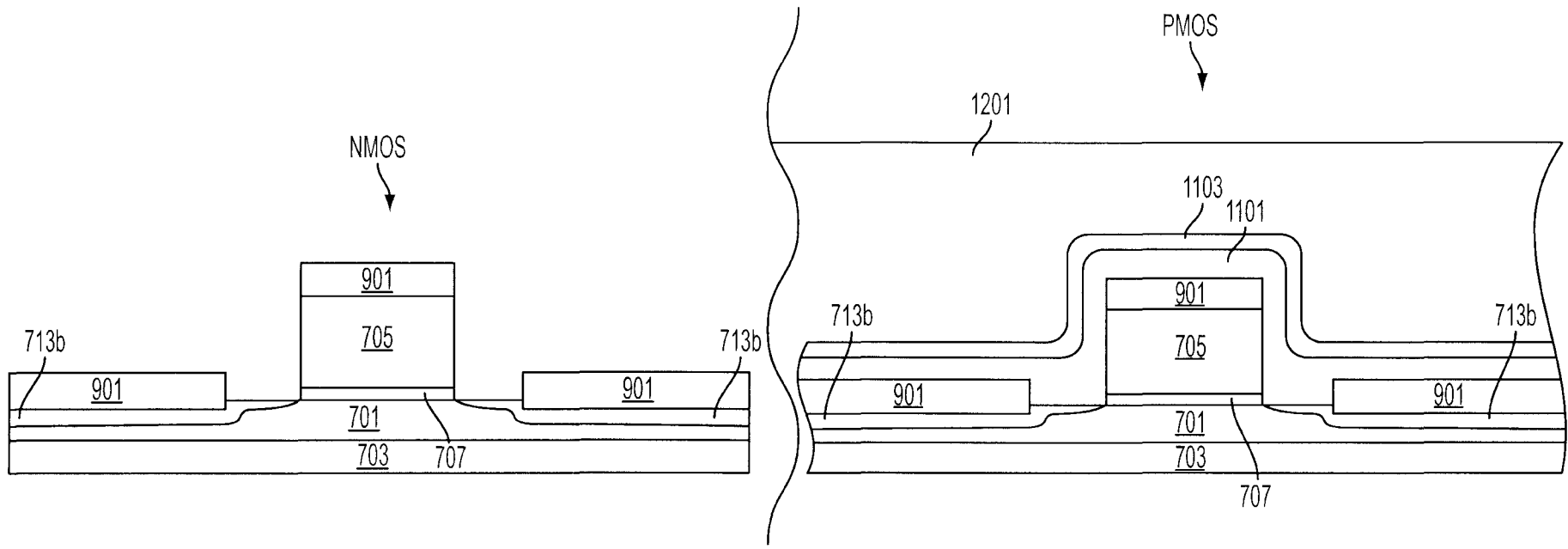


FIG. 12

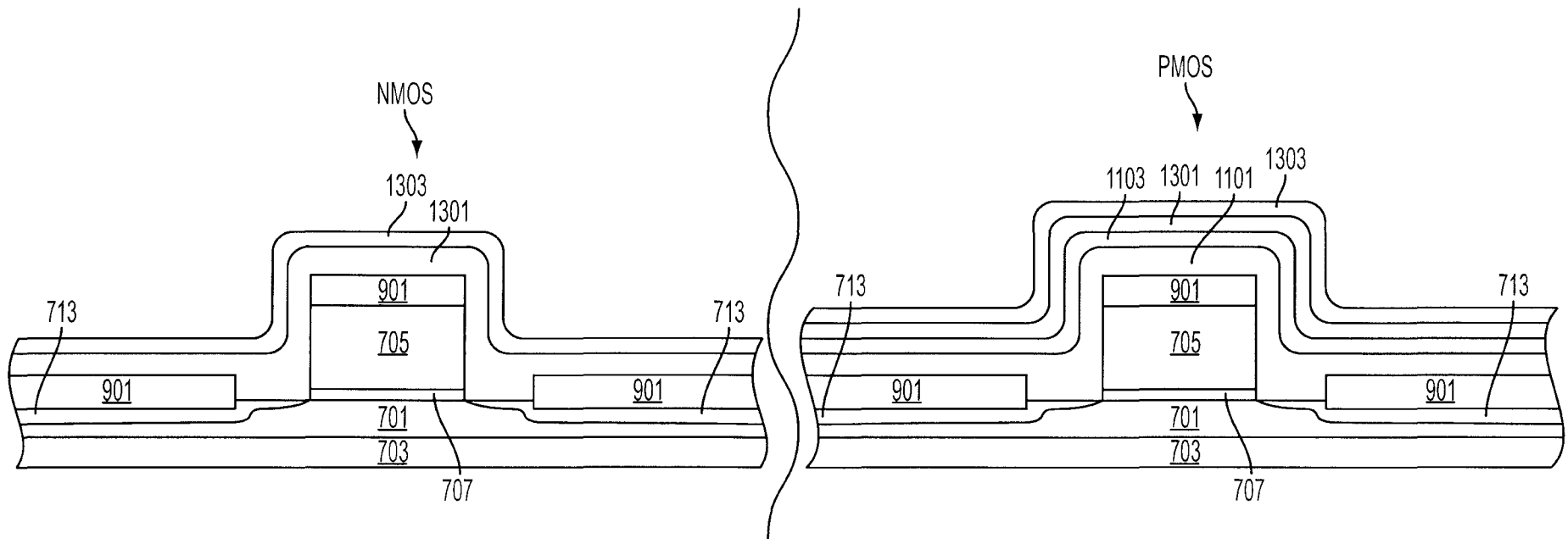


FIG. 13

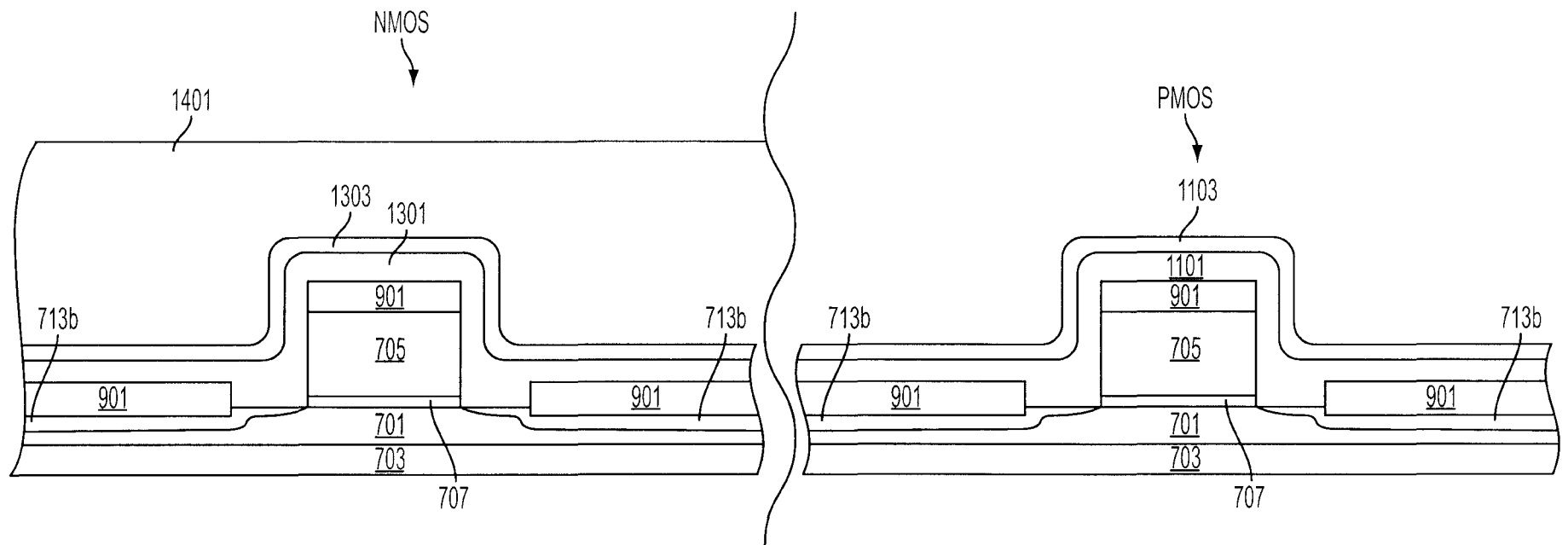


FIG. 14

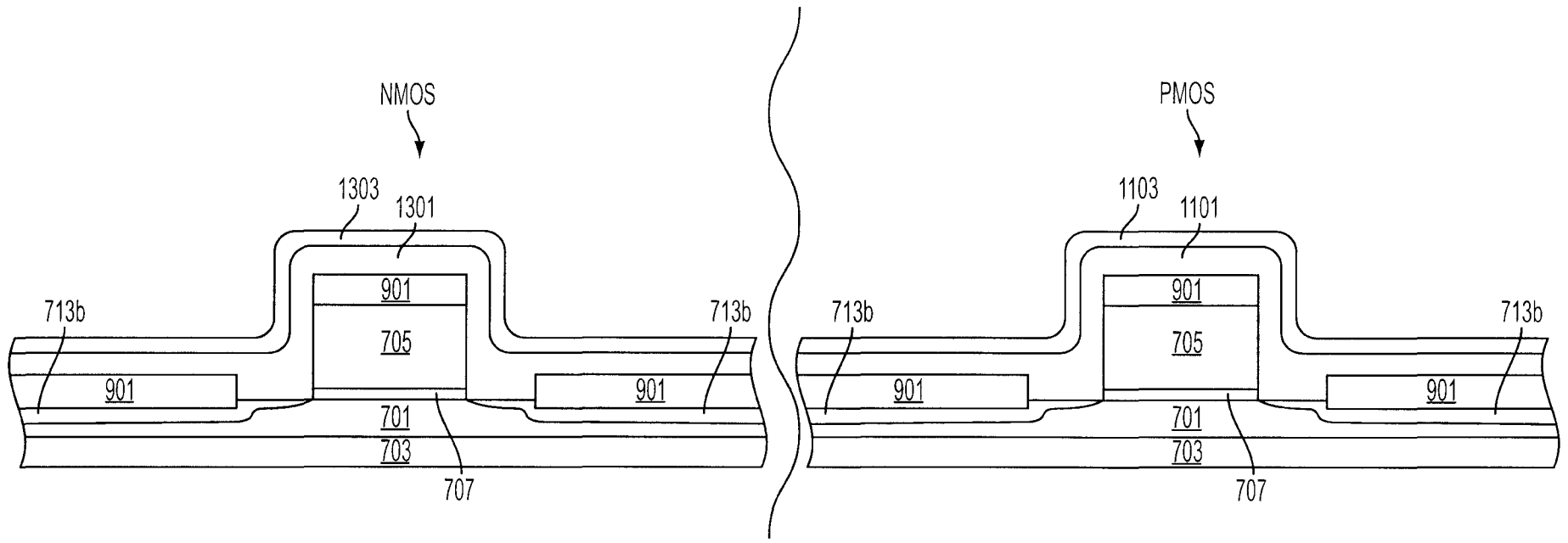


FIG. 15

Electronic Acknowledgement Receipt

EFS ID:	4949202
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	30671
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	01158-1030
Receipt Date:	11-MAR-2009
Filing Date:	05-FEB-2009
Time Stamp:	17:59:10
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Preliminary Amendment	NY0281_01158-1030_PreliminaryAmendment_filed03-11-09.pdf	22800 f996a289326052f62fc4be2898894ceae48b3426	no	3

Warnings:

Information:

2	Drawings-only black and white line drawings	NY0281_01158-1030_Replace mentDrawings_files03-11-09. pdf	364505 08141ba32b2be79fe3ffadbf9f39746c5655f 48a	no	13
Warnings:					
Information:					
Total Files Size (in bytes):				387305	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/366,378		Filing Date 02/05/2009		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)		SMALL ENTITY ☐		OR		OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A		N/A		N/A		N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A		N/A		N/A		N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A		N/A		N/A		N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$	=	X \$	=	X \$	=	X \$	=
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=	X \$	=	X \$	=	X \$	=
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)		(Column 3)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY	
AMENDMENT	03/11/2009	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)
Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$	=	X \$	=	X \$	=
Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	X \$	=	X \$	=	X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE		TOTAL ADD'L FEE		TOTAL ADD'L FEE	0
(Column 1)			(Column 2)		(Column 3)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY	
AMENDMENT	Total (37 CFR 1.16(i))	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)
Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	X \$	=	X \$	=
Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	X \$	=	X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE		TOTAL ADD'L FEE		TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/DONNA D. SMALLS LOGAN/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/366,378	02/05/2009	Karthik Ramani	01158-1030

CONFIRMATION NO. 7721

PUBLICATION NOTICE



OC000000042912746

30671
DITTHAVONG MORI & STEINER, P.C.
918 Prince Street
Alexandria, VA 22314

Title:METHOD TO REDUCE MOL DAMAGE ON NiSi

Publication No.US-2010-0193876-A1

Publication Date:08/05/2010

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/366,378	02/05/2009	Karthik Ramani	01158-1030	7721

99963 7590 02/10/2011
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT	PAPER NUMBER
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2813

NOTIFICATION DATE	DELIVERY MODE
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02/10/2011

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docket@dcpatent.com

Office Action Summary	Application No. 12/366,378	Applicant(s) RAMANI ET AL.	
	Examiner Joseph C. Nicely	Art Unit 2813	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 1 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 05 February 2009.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 1-20 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☐ Claim(s) _____ is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☐ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. _____ |
| 3) ☐ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date _____ | 6) ☐ Other: _____ |

DETAILED ACTION

Election/Restrictions

1. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims 1-13 and 20, drawn to a method, classified in class 438, subclass 592.
 - II. Claims 14-19, drawn to a device, classified in class 257, subclass 377.

The inventions are distinct, each from the other because of the following reasons:

2. Inventions I and II are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case the product as claimed can be made by another and materially different process. For instance, the product as claimed can be made by a process where a nickel silicide is formed first using a nickel layer deposited over the transistor and the platinum gradient can be formed after the silicide is formed through an ion implantation process.

3. Restriction for examination purposes as indicated is proper because all these inventions listed in this action are independent or distinct for the reasons given above and there would be a serious search and/or examination burden if restriction were not required because at least the following reason(s) apply:

the patentably distinct inventions require a different field of search (e.g., searching

Art Unit: 2813

different classes/subclasses and/or employing different search strategies or search queries).

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a invention to be examined even though the requirement may be traversed (37 CFR 1.143) and (ii) identification of the claims encompassing the elected invention.

The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other invention.

4. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be amended in compliance with 37 CFR 1.48(b) if one

Art Unit: 2813

or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. Any amendment of inventorship must be accompanied by a request under 37 CFR 1.48(b) and by the fee required under 37 CFR 1.17(i).

5. The examiner has required restriction between product and process claims.

Where applicant elects claims directed to the product, and the product claims are subsequently found allowable, withdrawn process claims that depend from or otherwise require all the limitations of the allowable product claim will be considered for rejoinder.

All claims directed to a nonelected process invention must require all the limitations of an allowable product claim for that process invention to be rejoined.

In the event of rejoinder, the requirement for restriction between the product claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product are found allowable, an otherwise proper restriction requirement between product claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product claim will not be rejoined. See MPEP § 821.04(b). Additionally, in order to retain the right to rejoinder in accordance with the above policy, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product claims. **Failure to do so may result in a loss of the right to rejoinder.** Further, note that the prohibition against double

Art Unit: 2813

patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Joseph C. Nicely whose telephone number is (571) 270-3834. The examiner can normally be reached on Monday through Friday 7:30AM-5:00PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Matthew Landau can be reached on (571) 272-1731. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2813

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Joseph C. Nicely/
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art
Unit 2813

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of: Karthik RAMANI <i>et al.</i>	Confirmation No.: 7721
Application No.: 12/366,378	Examiner: Nicely, Joseph C.
Filed: February 5, 2009	Group Art Unit: 2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

RESPONSE TO RESTRICTION REQUIREMENT

Dear Sir:

In response to the Restriction Requirement imposed in the Office Action dated February 10, 2011, Applicants elect to prosecute Group I, claims 1 through 13 and 20, drawn to a method. Favorable consideration of laser method claims 1 through 13 and 20 is solicited.

To the extent necessary, a petition for an extension of time under 37 C.F.R. 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,

DITTHAVONG MORI & STEINER, P.C.

February 15, 2011
Date

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106

918 Prince Street
Alexandria, VA 22314
Tel. (703) 519-9954
Fax. (703) 519-9958

Electronic Acknowledgement Receipt

EFS ID:	9444446
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	01158-1030
Receipt Date:	15-FEB-2011
Filing Date:	05-FEB-2009
Time Stamp:	13:52:18
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	NY0281_01158-1030_Response toRestrictionRequirement_filed 02-15-11.pdf	103218 080bf55ed95dba2f9c820e2fb564c9b78640dabc	no	1

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
12/366,378		2813	



Correspondence Address/Fee Address Change

The following fields have been set to Customer Number 99963 on 01/07/2011

- Correspondence Address
- Maintenance Fee Address
- Power of Attorney Address

The address of record for Customer Number 99963 is:

99963
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314



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NOTICE OF ALLOWANCE AND FEE(S) DUE

99963 7590 03/25/2011
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ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT

PAPER NUMBER

2813

DATE MAILED: 03/25/2011

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/366,378

02/05/2009

Karthik Ramani

01158-1030

7721

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$0	\$1810	06/27/2011

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
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Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

99963 7590 03/25/2011
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/366,378	02/05/2009	Karthik Ramani	01158-1030	7721
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TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional	NO	\$1510	\$300	\$0	\$1810	06/27/2011
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EXAMINER	ART UNIT	CLASS-SUBCLASS
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NICELY, JOSEPH C	2813	438-592000
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1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
- (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/366,378	02/05/2009	Karthik Ramani	01158-1030	7721

99963 7590 03/25/2011
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT	PAPER NUMBER
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2813

DATE MAILED: 03/25/2011

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 311 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 311 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability

Application No.

12/366,378

Examiner

Joseph C. Nicely

Applicant(s)

RAMANI ET AL.

Art Unit

2813

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the response to election/restriction filed 02/15/2011.
2. ☒ The allowed claim(s) is/are 1-13 and 20.
3. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).
- * Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

4. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
5. ☒ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
- (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
- 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
- (b) ☒ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date 20110318.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892)
2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)
3. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date 02/05/2009
4. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material
5. ☐ Notice of Informal Patent Application
6. ☒ Interview Summary (PTO-413), Paper No./Mail Date 20110317.
7. ☒ Examiner's Amendment/Comment
8. ☒ Examiner's Statement of Reasons for Allowance
9. ☐ Other ____.

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

<i>Examiner-Initiated Interview Summary</i>	Application No. 12/366,378	Applicant(s) RAMANI ET AL.	
	Examiner Joseph C. Nicely	Art Unit 2813	

All Participants:

(1) Joseph C. Nicely.

(2) Chris Ward (Reg. No. 41,367).

Date of Interview: 17 March 2011

Type of Interview:
☒ Telephonic
☐ Video Conference
☐ Personal (Copy given to: ☐ Applicant ☐ Applicant's representative)
Exhibit Shown or Demonstrated: ☐ Yes ☐ No
If Yes, provide a brief description: .

Part I.
Rejection(s) discussed:
N/A

Claims discussed:
2, 11, and 13

Prior art documents discussed:
N/A

Part II.
SUBSTANCE OF INTERVIEW DESCRIBING THE GENERAL NATURE OF WHAT WAS DISCUSSED:
See Continuation Sheet

Part III.
☒ It is not necessary for applicant to provide a separate record of the substance of the interview, since the interview directly resulted in the allowance of the application. The examiner will provide a written summary of the substance of the interview in the Notice of Allowability.
☐ It is not necessary for applicant to provide a separate record of the substance of the interview, since the interview did not result in resolution of all issues. A brief summary by the examiner appears in Part II above.

Status of Application: Allowable

(3) _____.

(4) _____.

Time: 4:30 p.m.

(Applicant/Applicant's Representative Signature – if appropriate)

Continuation of Substance of Interview including description of the general nature of what was discussed: Examiner contacted applicant's representative to obtain authorization to make changes to the aforementioned claims in an Examiner's Amendment to place the case into condition for allowance. Authorization was given. See attached Examiner's Amendment for details of the changes..

DETAILED ACTION

Election/Restrictions

1. Applicant's election of Group I (claims 1-13 and 20) in the reply filed on 02/15/2011 is acknowledged. Because applicant did not distinctly and specifically point out the supposed errors in the restriction requirement, the election has been treated as an election without traverse (MPEP § 818.03(a)).
2. Claims 14-19 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected Group, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 02/15/2011.
3. This application is in condition for allowance except for the presence of claims 14-19 directed to a group non-elected without traverse. Accordingly, claims 14-19 been cancelled.

Drawings

4. Figure 1 should be designated by a legend such as --Prior Art-- because only that which is old is illustrated. See MPEP § 608.02(g). Corrected drawings in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. The replacement sheet(s) should be labeled "Replacement Sheet" in the page header (as per 37 CFR 1.84(c)) so as not to obstruct any portion of the drawing figures. If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

EXAMINER'S AMENDMENT

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5. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in a telephone interview with Chris Ward (Reg. No. 41,367) on 03/17/2011.

The application has been amended as follows:

Claim 2, line 2, "of about from" is amended to read –from about–;

Claim 11, line 2, "first nickel layer" is amended to read –first layer–;

Claim 13, line 1, "to claim 1," is amended to read –to claim 12,–.

Cancel claims 14-19 as being drawn to a non-elected invention withdrawn without traverse.

Allowable Subject Matter

6. Claims 1-13 and 20 are allowed over the prior art of record.

7. The following is an examiner's statement of reasons for allowance: the prior art fails to teach in combination a method comprising forming a first layer of nickel, containing a first amount of platinum, over the transistor(s), forming a second layer of nickel, containing a second amount of platinum, on the first layer of nickel, wherein the second amount of platinum is greater than the first amount of platinum, and annealing the first and second layers of nickel to form a nickel silicide, wherein the nickel silicide layer has a platinum composition gradient with platinum increasing in amount in a

Art Unit: 2813

direction away from the source/drain regions and the gate electrode, and including all limitations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

8. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Arunachalam et al (US 2009/0127594) discloses forming two platinum containing layers over a transistor and annealing to form a silicide with a platinum gradient within the silicide. Arunachalam, however, discloses the opposite arrangement from the claimed invention, specifically forming the second platinum containing layer to have a lower concentration of platinum such that the silicide has a platinum concentration gradient decreasing away from the gate electrode. Arunachalam does not teach or suggest that reversing the ordering of the formation of the platinum containing layers to obtain the claimed arrangement would result in the same benefits of the ordering/structuring shown by Arunachalam.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Joseph C. Nicely whose telephone number is (571) 270-3834. The examiner can normally be reached on Monday through Friday 8:30AM-6:00PM EST.

Art Unit: 2813

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Matthew Landau can be reached on (571) 272-1731. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Joseph C. Nicely/
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art
Unit 2813

Notice of References Cited	Application/Control No. 12/366,378		Applicant(s)/Patent Under Reexamination RAMANI ET AL.	
	Examiner Joseph C. Nicely		Art Unit 2813	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2005/0247926 A1	11-2005	Sun et al.	257/019
*	B	US-2006/0051961 A1	03-2006	Cabral et al.	438/664
*	C	US-2006/0284263 A1	12-2006	Chang et al.	257/383
*	D	US-2007/0093047 A1	04-2007	Okuno et al.	438/592
*	E	US-2009/0127594 A1	05-2009	ARUNACHALAM et al.	257/288
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 12366378	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
	Examiner Joseph C Nicely	Art Unit 2813

SEARCHED			
Class	Subclass	Date	Examiner
438	197, 199, 275, 592, 651, 655, 664, 682	3/4/2011	JCN
257	E21.203, E21.296, E21.438, E21.622, E21.636	3/4/2011	JCN

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Search (see attached history)	3/4/2011	JCN
Inventor Name Search conducted through PALM and EAST	3/4/2011	JCN
Discussed claim interpretation, prior art, prior art search, and allowable subject matter with SPE Matt Landau	3/8/2011	JCN
STIC search conducted by Scott Segal	3/15/2011	JCN

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
438	592, 651, 682	3/18/2011	JCN
257	E21.636	3/18/2011	JCN

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	21	(platinum OR Pt).clm. WITH (concentration OR gradient). clm. WITH (NiSi OR silicide). clm.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/18 09:51
S1	2	((("6841449") or ("20050247926")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/03/04 11:22
S2	16	("2005/0247926").URPN.	USPAT	ADJ	ON	2011/03/04 11:23
S3	14	S2 AND (silicide OR suicide OR silicid\$8)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:23
S4	0	("6841449").URPN.	USPAT	ADJ	ON	2011/03/04 11:25
S5	1	("6521529").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2011/03/04 11:25
S6	20	("20010053426" "20020000622" "5434109" "5759906" "5804492" "6001721" "6046089" "6074921" "6127238" "6129819" "6165915" "6211035" "6217658" "6218251" "6218275" "6228730" "6245669" "6268297" "6323561" "6326263").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2011/03/04 11:25
S7	7	("6521529").URPN.	USPAT	ADJ	ON	2011/03/04 11:25
S8	5825	(nickel OR Ni) WITH (silicide OR suicide OR salicide OR silicid\$7 OR suicid\$7 OR salicid\$7) WITH (platinum OR Pt)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:26
S9	4137	(nickel OR Ni) NEAR4 (platinum OR Pt) WITH (silicide OR suicide OR salicide OR silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:27

S10	847	(nickel platinum OR NiPt) WITH (silicide OR suicide OR salicide OR silicid\$7 OR suicid \$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:27
S11	795	S10 and (@pd<"20090205" or @ad<"20090205" or @prad<"20090205" or @rlad<"20090205")	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:28
S12	298	(ramani near karthik).in. OR (besser NEAR paul).in.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:32
S13	25	S12 and (nickel OR Ni) NEAR4 (platinum OR Pt) WITH (silicide OR suicide OR salicide OR silicid\$7 OR suicid \$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 11:32
S14	4	("20070004205" "20070093047" "20070004205" "20070093047" "20090028744" "20090127594").PN.	US-PGPUB; USPAT; FPRS; EPO; JPO; IBM_TDB	ADJ	ON	2011/03/04 11:35
S15	1	("20070093047").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/03/04 12:01
S16	7	("20040097060" "20040209432" "20050158996" "20060051961" "6251779" "6335253" "6936528"). PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2011/03/04 12:34
S17	0	("Ni.sub.1-x" "Pt.sub.x") WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:39
S18	0	("Ni.sub.1-xPt.sub.x") WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:40
S19	0	("Ni.sub.1-x Pt.sub.x") WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:40

S20	0	("Ni.sub.x Pt.sub.1-x") WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:40
S21	12	((first OR lower oR great\$4) NEAR3 (NiPt OR nickel platinum)) SAME ((second OR upper oR less\$4) NEAR3 (NiPt OR nickel platinum))	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:41
S22	251	(first WITH (NiPt OR nickel platinum)) SAME (second WITH (NiPt OR nickel platinum))	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:42
S23	64	(first WITH (NiPt OR nickel platinum)) SAME (second WITH (NiPt OR nickel platinum)) SAME (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:43
S24	9	(Pt OR platinum) NEAR3 (gradient) WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:47
S25	3	(Pt OR platinum) NEAR3 (var\$8 OR chang\$7 OR decreas\$7 OR increas\$7) NEAR3 (composit\$8 OR percent\$7 OR amount OR weight) WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:49
S26	2619	(NiPtSi OR NiPt NEAR2 (silicid\$7 OR suicid\$7 OR salicid\$7)) OR ((nickel OR Ni) NEAR3 (silicid\$7 OR suicid\$7 OR salicid\$7) WITH (Pt OR platinum))	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:51
S27	179	(NiPtSi OR NiPt NEAR2 (silicid\$7 OR suicid\$7 OR salicid\$7)) OR ((nickel OR Ni) NEAR3 (silicid\$7 OR suicid\$7 OR salicid\$7) WITH (Pt OR platinum) NEAR2 (gradient OR var\$7 OR chang\$7 OR decreas\$7 OR increas\$7 OR less\$7 OR great\$7))	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:51
S28	161	S27 and (@pd<"20090205" or @ad<"20090205" or @prad<"20090205" or @rlad<"20090205")	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 12:52

S29	14	(silicid\$7 OR suicid\$7 OR salicid\$7) WITH (Pt OR platinum) WITH (proportion)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:02
S30	1	(12/366378).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/03/04 15:06
S31	3195	438/197.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:08
S32	148	S31 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:09
S33	2185	438/199.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:15
S34	1927	S33 NOT S31	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:15
S35	105	S34 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:15
S36	3483	438/275,592.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:22
S37	3097	S36 NOT (S31 or S33)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:23
S38	187	S37 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:23
S39	2089	438/651,655.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:31

S40	1747	S39 NOT (S31 or S33 or S36)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:31
S41	166	S40 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:31
S42	1532	438/664,682.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:36
S43	838	S42 NOT (S31 or S33 or S36 or S39)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:36
S44	164	S43 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:36
S45	2927	257/E21.164,E21.165, E21.199.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:45
S46	1646	S45 NOT (S31 or S33 or S36 or S39 or S42)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:45
S47	246	S46 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:45
S48	1	("20060068546").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/03/04 15:49
S49	3749	257/E21.203,E21.296, E21.438.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:53
S50	1890	S49 NOT (S31 or S33 or S36 or S39 or S42 or S45)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:54

S51	257	S50 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 15:54
S52	1020	257/E21.622,E21.636.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 16:02
S53	297	S52 NOT (S31 or S33 or S36 or S39 or S42 or S45 or S49)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 16:02
S54	37	S53 AND (NiPt OR nickel platinum OR platinum) WITH (salicid\$7 OR silicid\$7 OR suicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 16:02
S55	224	(Pt OR platinum) NEAR4 (gradient OR composition\$4 OR var\$7 OR chang\$6 OR decreas\$6 OR increas\$6) WITH (silicid\$7 OR suicid\$7 OR salicid\$7)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 16:05
S56	220	S55 and (@pd<"20090205" or @ad<"20090205" or @prad<"20090205" or @rlad<"20090205")	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/04 16:05

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L2	21	(platinum OR Pt).clm. WITH (concentration OR gradient).clm. WITH (NiSi OR silicide).clm.	US-PGPUB; USPAT; UPAD	ADJ	ON	2011/03/18 09:52

3/ 18/ 2011 10:29:21 AM**H:\ EAST Workspaces\ FAOM\ 12366378.w sp**

<i>Index of Claims</i> 	Application/Control No. 12366378	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
	Examiner Joseph C Nicely	Art Unit 2813

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	03/18/2011								
1	1	=								
2	2	=								
3	3	=								
4	4	=								
5	5	=								
6	6	=								
7	7	=								
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	18	-								
	19	-								
14	20	=								



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BIB DATA SHEET

CONFIRMATION NO. 7721

SERIAL NUMBER 12/366,378	FILING or 371(c) DATE 02/05/2009 RULE	CLASS 257	GROUP ART UNIT 2813	ATTORNEY DOCKET NO. 01158-1030		
APPLICANTS Karthik Ramani, Hopewell Junction, NY; Paul R. Besser, Sunnyvale, CA; ** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 02/19/2009						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/JOSEPH C. NICELY/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY NY	SHEETS DRAWINGS 13	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314 UNITED STATES						
TITLE METHOD TO REDUCE MOL DAMAGE ON NiSi						
FILING FEE RECEIVED 1090	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees			
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			☐ Other _____			
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Receipt date: 02/05/2009

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-09)

Approved for use through 02/28/2009. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		12/366,378	
	Filing Date		2009-02-05	
	First Named Inventor	Karthik RAMANI et al.		
	Art Unit	2813		
	Examiner Name	J. Nicely		
	Attorney Docket Number	01158-1030		

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/J.C.N./	1	6841449		2005-01-11	Bertrand et al.		

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/J.C.N./	1	20050247926	A1	2005-11-10	Sun et al.		

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Receipt date: 02/05/2009 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		12/366,378
	Filing Date		2009-02-05
	First Named Inventor	Karthik RAMANI et al.	
	Art Unit	2813	
	Examiner Name	J. Nicely	
	Attorney Docket Number	01158-1030	

	1		☐
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EXAMINER SIGNATURE

Examiner Signature	/Joseph C. Nicely/	Date Considered	03/04/2011
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of: Karthik RAMANI <i>et al.</i>	Confirmation No.: 7721
Application No.: 12/366,378	Examiner: Nicely, Joseph C.
Filed: February 5, 2009	Group Art Unit: 2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

RESPONSE UNDER 37 CFR §1.312

Dear Sir:

In response to the Notice of Allowance dated March 25, 2011, please amend the above-identified patent application prior to issuance as follows:

AMENDMENT TO THE DRAWINGS	2
REMARKS	3

AMENDMENT TO THE DRAWINGS

The attached Replacement Sheet of drawings includes changes to FIG. 1. This Replacement Sheet, which includes FIG. 1, replaces the original sheet including FIG. 1.

Attachment: Replacement Sheet (1)

REMARKS

Favorable reconsideration of this application in light of the following discussion is respectfully requested.

Claims 1-13 and 20 are presently pending in the present application.

The Applicants are submitting concurrently herewith an amendment to FIG. 1 that adds the legend “(PRIOR ART)”, as requested in the Notice of Allowance. Accordingly, the Applicants request entry of the amendment and withdrawal of the drawing objection.

Therefore, the present application is in condition for allowance. Favorable consideration is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9957 so that such issues may be resolved as expeditiously as possible.

To the extent necessary, a petition for an extension of time under 37 CFR §1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,

DITTHAVONG MORI & STEINER, P.C.

April 15, 2011
Date

/Christopher D. Ward/
Christopher D. Ward
Attorney/Agent for Applicant(s)
Reg. No. 41,367

918 Prince Street
Alexandria, VA 22314
Tel. (703) 519-9951
Fax. (703) 519-9958

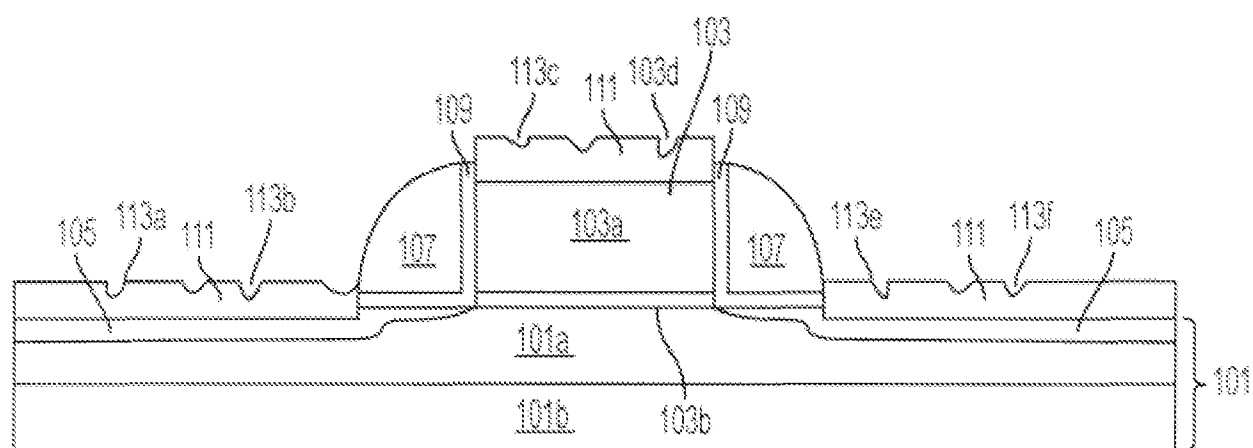


FIG. 1
(PRIOR ART)

Electronic Acknowledgement Receipt

EFS ID:	9891182
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NISI
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Christopher D. Ward/Brandy Stoeckel
Filer Authorized By:	Christopher D. Ward
Attorney Docket Number:	01158-1030
Receipt Date:	15-APR-2011
Filing Date:	05-FEB-2009
Time Stamp:	16:55:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NY0218_01158-1030_Rule312A mendment_filed04-15-11.pdf	143711 3f31d3396a02dd4851ce5dbd9b27367c3167fccd	yes	4

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment after Notice of Allowance (Rule 312)	1	1
	Drawings-only black and white line drawings	2	2
	Applicant Arguments/Remarks Made in an Amendment	3	3
	Drawings-only black and white line drawings	4	4

Warnings:

Information:

Total Files Size (in bytes):	143711
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B – FEE(S) TRANSMITTAL

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Commissioner for Patents
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Alexandria, Virginia 22313-1450
or Fax (571) 273-2885**

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(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/366,378	February 5, 2009	Karthik Ramani	01158-1030	7721

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NiSi

APPLN. TYPE	SMALL ENTITY	ISSUE FEE	PUBLICATION FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$1810	06/27/2011

EXAMINER	ART UNIT	CLASS-SUBCLASS
NICELY, JOSEPH C	2813	438-592000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

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(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Ditthavong Mori & Steiner, P.C.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

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GLOBALFOUNDRIES Inc. GRAND CAYMAN, CAYMAN ISLANDS

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

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- ☐ The Director is hereby authorized by charge the required fee(s), or credit any overpayment, to Deposit Account Number _____.

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27 (g)(2).

The Directory of the USPTO is requested to apply the Issue Fee and Publication Fee (if any) or to re-apply any previously paid issue fee to the application identified above. NOTE: The Issue Fee and publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Christopher D. Ward/

Date June 24, 2011

Typed or printed name Christopher D. Ward

Registration No. 41367

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Electronic Patent Application Fee Transmittal

Application Number:	12366378			
Filing Date:	05-Feb-2009			
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NISI			
First Named Inventor/Applicant Name:	Karthik Ramani			
Filer:	Arthur J. Steiner/Brandy Stoeckel			
Attorney Docket Number:	01158-1030			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1510	1510
Publ. Fee- early, voluntary, or normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1810

Electronic Acknowledgement Receipt

EFS ID:	10378130
Application Number:	12366378
International Application Number:	
Confirmation Number:	7721
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NISI
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	01158-1030
Receipt Date:	24-JUN-2011
Filing Date:	05-FEB-2009
Time Stamp:	13:43:08
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1810
RAM confirmation Number	12101
Deposit Account	504213
Authorized User	DITTHAVONG,PHOUPHANOMKETH

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Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	NY0281_01158-1030_PartBFee Transmittal_filed06-24-11.pdf	106780 5fe732cf3da1cc711806454c93a06a5a30bd8831	no	1

Warnings:**Information:**

2	Fee Worksheet (SB06)	fee-info.pdf	32172 07fdd0feb33778c912a12a4716f44d8350aa0f2d	no	2
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Warnings:**Information:**

Total Files Size (in bytes):			138952
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	7721
Application No.:	12/366,378	Examiner:	Nicely, Joseph C.
Filed:	February 5, 2009	Group Art Unit:	2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

RESPONSE UNDER 37 CFR §1.312

Dear Sir:

In response to the Notice of Allowance dated March 25, 2011, please amend the above-identified patent application prior to issuance as follows:

AMENDMENT TO THE DRAWINGS 2

REMARKS 3



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/366,378	02/05/2009	Karthik Ramani	01158-1030	7721

99963 7590 07/08/2011
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT	PAPER NUMBER
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2813

NOTIFICATION DATE	DELIVERY MODE
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07/08/2011

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docket@dcpatent.com

<i>Response to Rule 312 Communication</i>	Application No.	Applicant(s)
	12/366,378	RAMANI ET AL.
	Examiner	Art Unit
	Joseph C. Nicely	2813

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

1. ☒ The amendment filed on 15 April 2011 under 37 CFR 1.312 has been considered, and has been:

a) ☒ entered.

b) ☐ entered as directed to matters of form not affecting the scope of the invention.

c) ☐ disapproved because the amendment was filed after the payment of the issue fee.

Any amendment filed after the date the issue fee is paid must be accompanied by a petition under 37 CFR 1.313(c)(1) and the required fee to withdraw the application from issue.

d) ☐ disapproved. See explanation below.

e) ☐ entered in part. See explanation below.

The replacement Figure 1 overcomes the objection to the drawing raised in the Notice of Allowance mailed 03/25/2011 and is thus entered.

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

/Joseph C. Nicely/
Examiner, Art Unit 2813



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/366,378	08/09/2011	7994038	01158-1030	7721

99963 7590 07/20/2011
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 194 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Karthik Ramani, Hopewell Junction, NY;
Paul R. Besser, Sunnyvale, CA;